European Commission

Joint Research Centre

Blue Mirror

One Water - One Health Monthly Chronicle



SEPTEMBER 2025
VOL 1 ISSUE 1

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JRC144308

PDF ISSN 3094-5836 KJ-01-25-554-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: Gawlik, B.M., Panzarella, G., Comero, S., Tessarolo, A., Koutsolioutsou, A. et al., *Blue Mirror - One Water - One Health Monthly Chronicle - Vol. 1 Issue 1 - September 2025*, Publications Office of the European Union, Luxembourg, 2025, JRC144308.

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Cover Image
Image created by JRC, with minor modifications made using *Photoshop*. The allegorical image shows a woman walking up a staircase. Each step bears an abstract, artistic representation of EU and non-EU icons. The staircase rises from the water toward a light above, symbolising progress from shared challenges in wastewater pathogen surveillance to stronger public health monitoring and decision-making.

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EDITORIAL
Blue Mirror, First Edition

Water embodies the essence of our societies, serving as a profound *mirror* that unveils the intricate patterns of health, behaviour, and resilience.
What emerges from this reflection transcends mere data; it becomes a compass, an imperative guide directing us toward policies that nurture well-being and justice within our communities.
Blue Mirror represents the natural progression of the EU Wastewater for Public Health bulletin into a global journal focused on delivering timely, rigorous, and policy-relevant insights drawn from wastewater pathogen surveillance.
Our commitment remains steadfast: to provide information grounded in data from European dashboards and official sources, integrated through the EU Wastewater Observatory, and guided by wastewater-based epidemiology (WBE) as an essential instrument for public health intelligence.
The framework is evolving to weave together scientific rigour, informed decision-making, and preparedness for pathogens such as SARS-CoV-2, avian influenza, mpox, RSV, and seasonal influenza. It will emphasise not only the policy relevance and the crucial role of stakeholder but also, the importance of cooperation and the insights gained from lived experience.

In our discipline, water acts as a living archive, a dynamic "vellum" that continuously inscribes the biological and chemical signatures of populations.
Interpreting this liquid parchment with precision reveals outbreaks, tracks antimicrobial resistance, and uncovers public health patterns often invisible to conventional systems. Yet to unlock these benefits, we must rely on rigorous analytical methods, interoperable data platforms, and responsible stewardship to preserve the integrity and meaning of the data.

Nietzsche's insight: "the most valuable truths are discovered last; but the most valuable truths are methods", reminds us that knowledge is not secured merely by what is seen, but by the lens through which seeing becomes possible.

The worth of our insights lies in the rigour and coherence of the methods that generate them, for method is the silent architecture of discovery. Within the domain of WBE, this methodological integrity is the vital condition for producing knowledge that does not merely illuminate the present but endures as a guide for action.
Such integrity can only be sustained through a continual dialogue across disciplines and borders, where empirical evidence and lived experience converge to shape policies that are both just and effective.

The staircase depicted on our cover symbolises this continuous ascent - a collective and international journey toward a healthier, more equitable future. Step by step, we delve beneath the surface to uncover hidden vulnerabilities and detect signals imperceptible from above, translating them into knowledge that safeguards society. Through systematic sample collection, standardised laboratory processing, and rigorously validated detection protocols, this surveillance paradigm captures population-level disease indicators that often precede clinical case identification by several days.

The methodology's remarkable sensitivity to detect viral RNA or other pathogenic indicators allows for early warning systems that inform evidence-based public health interventions, providing epidemiological insights that complement traditional disease surveillance networks. In June

EDITORIAL
Blue Mirror, First Edition

2025, the European Commission's Health Emergency Preparedness and Response Authority (HERA), in partnership with the Joint Research Centre (JRC), launched the sample collection phase of the European Super-Sites initiative.

This initiative, conducted under the HaDEA framework contract, aims to enhance early warning capabilities and coordination near strategic transport hubs.

By translating environmental signals into timely public health responses, this program strengthens both European and global preparedness.

In alignment with this mission, Blue Mirror introduces a dedicated column: EU Super-Sites. Insights from Outsight: which offers a summary of the wastewater activities from strategic monitoring sites (e.g., airports, train stations, hospitals).

The Super-Sites operate within GLOWACON, the Global Consortium for Wastewater and Environmental Surveillance for Public Health, whose goal is to establish a worldwide sentinel system.

Each month, Blue Mirror will feature updates in News from GLOWACON: showcasing the consortium's most significant contributions to global health security.

This issue also inaugurates the Pathogen-Specific Reports section. This month, Blue Mirror focuses on two critical public health threats: West Nile virus and Chikungunya vector-borne diseases. Early detection and ongoing monitoring of these diseases substantially benefit from wastewater-based approaches, particularly when integrated with complementary surveillance systems.

Finally, we introduce three other columns:

1. *Correspondance*: this section welcomes feedback and reflections from experts, researchers, stakeholders, readers. It serves as an open platform to discuss the challenges, opportunities, and innovations in wastewater surveillance, public health preparedness, eco-health systems as a whole. Contributions may include expert opinions, community perspectives, and policy recommendations to further improve the effectiveness of early warning systems.
2. *Comment*: it is a dedicated space to share opinions, innovations, and latest findings from a wide range of fields including science, arts, culture, and policy. It provides a forum to reflect on recent discoveries, developments that impact public health, environmental surveillance, and society at large. Contributions may include interpretative insights, interdisciplinary perspectives and critical discussions aiming to inspire dialogue and advance knowledge across sectors.
3. *The Human Sample*: it is a dedicated space for contributions from beneficiaries, experts, and stakeholders. This section offers qualitative, societally relevant perspectives in the form of testimonials, highlighting the practical impact of wastewater surveillance data.

Blue Mirror aspires to be a dynamic platform where science informs policy and policy supports science, with the shared objective of protecting public health as a universal right.

We invite you to engage with this work, to reflect on the evidence, to question and to contribute to a dialogue that is as vital as the waters we study.

Welcome to Blue Mirror.

Sincerely,



Bernd Manfred Gawlik
Chief Editor

CORRESPONDANCE

A Call for «Multilogues»

Bernd Manfred Gawlik, Giulia Panzarella

The Correspondence section of Blue Mirror occupies a vital role in furthering the journal's mission to be a global platform for timely, rigorous, and policy-relevant insights drawn from water surveillance. Evolving naturally from the *EU Wastewater for Public Health* bulletin, Blue Mirror reflects the expanding scope of environmental and public health monitoring through water systems, encompassing not only wastewater pathogen surveillance but also broader aspects of water quality, ecosystem health, circular economy, related technological and policy developments.

In this interdisciplinary field, where environmental science, public health, arts, engineering, resource management, data science, and policy intersect, professional correspondence is an indispensable tool for fostering meaningful dialogue.

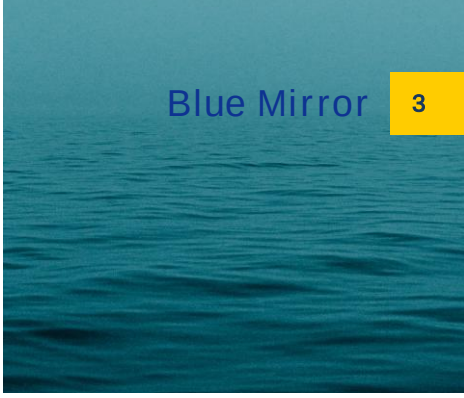
It offers a forum for critical evaluation of practices, sharing of innovative data and approaches, and thoughtful discussion of regulatory and societal implications. These exchanges enrich the collective understanding, encouraging collaboration across regions and sectors and inviting diverse viewpoints.

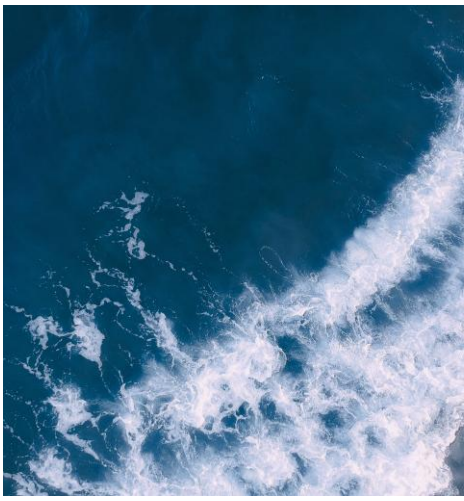
evidence-based policies and fostering resilient water management practices worldwide.

We would like this place to enable critical engagement, providing authors and readers the opportunity to propose alternative interpretations and viewpoints, or highlight emerging perspectives connected to published work.

We believe maintaining an open channel for correspondence is essential in upholding the journal's commitment to intellectual growth, being responsive to the evolving needs and inquiries of the community, promoting a culture of inclusivity and continual learning.

We therefore warmly invite researchers, practitioners, policymakers, readers, and all engaged stakeholders to contribute their perspectives, experiences, and critiques. Please submit your contributions to our editorial email team including [BLUEMIRROR] explicitly in the email subject.











COMMENT

Nobel Wisdom: Cooperation and Competition in Contemporary Science

Giulia Panzarella

Contemporary science operates on a razor's edge: on one side, the urgent need to excel in strategic sectors; on the other, the increasingly evident necessity to cooperate on objectives that transcend national borders and interests. "*Scientific knowledge is global capital that doesn't diminish when shared*", explains Brian Schmidt, Nobel laureate who spoke at Lindau in late August at the Nobel laureate meeting (he won the Physics Prize in 2011 for the discovery of the accelerating expansion of the Universe through observations of distant supernovae). "*If we want to advance, we must keep the channels of basic research permeable, publish with transparency, and ensure that results can take root in different places, because isolation leads to slower and more expensive solutions.*"

Today's configuration is tripolar: the United States, China, and Europe are pushing hard on research and industrialisation, but the speed of transfer from laboratory to market is not uniform. The Old Continent ranks third. History reminds us that trajectories intertwine: a century ago, the United States transformed European discoveries into mass products; today China follows a similar path but with a substantial difference, as it invests directly in research, energy, and advanced computing.

"*Competition is inevitable and, up to a point, healthy,*" adds Steven Chu, Stanford physics professor and 1997 Nobel laureate, who joined the economic sciences Nobel dialogue this year. "*The foundation of the next wave of innovations is built on open data, international collaborations, and reproducible results, not on rigid borders and mutual suspicion.*"

Within this framework, concrete decisions must be made.

The potential in managing wastewater demands a leap forward in global network management, where cooperation is not an act of generosity but a driver of efficiency. GLOWACON, the Global Consortium for Wastewater and Environmental Surveillance for Public Health was founded precisely on this principle. Sharing best practices reduces errors, reshapes incentives, and enables more countries to adopt robust solutions. "*Not everything is a zero-sum game; efficiency in management and digital integration can be areas where protocols, metrics, and standards are shared, so that one's progress becomes the asset of many.*" The same logic applies to innovation ecosystems: generating knowledge is not enough without agile bridges to entrepreneurship, incubators connected to public labs, and capital able to support ideas while they remain technical risks, before economic ones.

The most successful regions are characterised by close geographical and cultural ties among universities, businesses, and research centres, whereas in areas dominated by fragmentation and strict intellectual property protections, projects often fail before they even get off the ground.

While commercial competition drives quality and cost reduction, scientific progress demands a common language, because experiment reliability depends on reproducibility and shared methods, not secrecy.

In climate policy, this distinction is even clearer: credible decarbonisation targets require institutional architectures allowing groups of countries to act first by setting standards, coordinating investments, and opening financing channels where emission reductions are most cost-effective.

COMMENT

Nobel Wisdom: Cooperation and Competition in Contemporary Science

Figure A| A multimedia installation at Rio de Janeiro's Museu do Amanhã displays the message "Life is cooperation and competition," visually embodying evolution and the fundamental tension that drives both biological and scientific progress.



Source: JRC

"*Waiting for everyone to move often means not moving at all,*" Chu adds.

"*If a substantial bloc of countries converges on common rules and compatible tools first, others will naturally align later, once benefits are measurable and the learning curve is paid.*"

Even space activities highlight how fragile the boundary between opportunity and risk is. Astronomy and the International Space Station have shown that collaboration is possible, but low Earth orbit is becoming a congested common good: launching thousands of satellites without a shared framework creates interference, increases debris, and raises mitigation costs borne by all. "*Without global regulation, we risk turning a precious infrastructure into an orbital dump,*" warns Chu. "*We need access criteria, transparency on trajectories, and accountability for those who break the rules, or else innovation will turn into a brake on science.*"

Culturally, transdisciplinary exchange is indispensable: economic models, despite their limits, guide political decisions; natural sciences define the physical boundaries; and computer science opens new possibilities. We are all living a particular historical period, where democracy is declining, geopolitical tensions are growing, and artificial intelligence is likely profoundly affecting decision-making. How do we improve collective decision-making across cultures, geographies, and generations? Through methods. "Explaining the 'method' is part of scientific responsibility, because trust arises when people understand not only what we know but also what we do not yet know," Schmidt emphasises and continues:

"Cooperation accelerates the solution of common problems, competition improves quality *and accessibility, but it's fundamental to remain open in basic science, because that's where we prepare the discoveries that will one day become global public goods, regardless of where they originate*". Donella Meadows, a pioneering environmental scientist and systems thinker, would undoubtedly add a vital element: imagination. True progress hinges on our ability to envision a better world - its materials, technologies, systems, and models. As she eloquently urged, "*Close your eyes and imagine the sustainable world you desire.*"

COMMENT

The High Seas Treaty: From Genetic Resources to Planetary Stewardship

Giulia Panzarella

On September 19, 2025, Morocco and Sierra Leone became the 60th and 61st countries to sign the United Nations Agreement on the Conservation and Sustainable Use of Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ Agreement), commonly known as the *High Seas Treaty*. This landmark legally binding instrument under the United Nations Convention on the Law of the Sea (UNCLOS) represents the culmination of over two decades of international negotiations and diplomatic efforts. These critical signatures provided the necessary momentum for treaty approval, with the agreement set to enter into force on January 17, 2026.

The treaty's implementation represents a significant milestone in achieving the 2030 Agenda objectives, particularly the protection of 30% of marine and terrestrial areas, a target essential for maintaining planetary biodiversity and ecosystem integrity. A core element of the treaty is its comprehensive approach to marine genetic resources, which include the genetic material of marine organisms such as bacteria, corals, algae, and other ocean life forms. These genetic materials hold immense potential for scientific advancement and applications in biotechnology, pharmaceuticals, and other innovative fields. A particularly novel aspect is the regulation of Digital Sequence Information (DSI) - the genetic data derived from marine organisms - which will enhance transparency and traceability of genetic resource use. Marine biologist Chris Bowler from CNRS in France and member of the *Accademia dei Lincei* explains: "The agreement establishes a governance framework for the collaborative work of marine scientists, philanthropic organisations, and private companies engaged in resource exploration in international waters. It encourages voluntary declarations of research and exploration activities, ensuring thorough documentation of discoveries. The collected digital sequence information (DSI) facilitates benefit-sharing and capacity building, particularly benefiting developing nations, under the oversight of a newly formed United Nations management authority. This authority encompasses a Conference of the Parties (COP), scientific advisory committees, and regulatory mechanisms designed to ensure traceability and responsible utilisation of marine genetic resources."

The treaty also prioritises cross-border cooperation to address climate change impacts on marine ecosystems.

Climate scientist Sandro Carniel highlights the importance of mitigation over adaptation: "Mitigation involves proactive technological development to address root causes, such as reducing greenhouse gas emissions to manageable concentrations. These measures operate over extended timeframes and require sustained investment in innovative solutions. Adaptation, conversely, involves working with existing tools and resources to address immediate challenges without the luxury of developing new technologies. As Carniel explains, "In adaptation scenarios, we must work with what we have available, while others invest in long-term mitigation strategies".

To truly preserve the ocean's potential, we must favour mitigation through coordinated international efforts while simultaneously improving our capacity for adaptation to inevitable climate risks, including floods, droughts, and pollution.

COMMENT

The High Seas Treaty: From Genetic Resources to Planetary Stewardship

Our planet operates as an interconnected system where diverse flows create the conditions for life: the gradual drift of continents, swift ocean currents, rapid atmospheric winds, and the constant energy of solar radiation.
These dynamic processes form four fundamental "oceans“ - light, air, water, and land - that constitute and surround all terrestrial life. This holistic systems perspective is vividly illustrated in a captivating video (Video 1) from Rio de Janeiro's Museu do Amanhã (in English: Museum of Tomorrow), a groundbreaking exhibition that runs presently. The museum demonstrates how art, science, and innovation can converge around sustainability and coexistence values. Such interdisciplinary approaches inspire new perspectives for the world we seek to build. Science fundamentally requires imagination alongside rational analysis. The common misconception that science is purely rational, while imagination belongs exclusively to the arts, misrepresents the creative nature of scientific inquiry. Authentic scientific practice involves intuition, observation, creation.

As Donella Meadows emphasised in her book “Limit to Growth”, very often people think that science is something exclusively rational, so they associate imagination exclusively with the arts. This is incorrect. Doing science means imagining new scenarios while the world is engaged in its dance (Figure B). We must tread lightly on this planet, moving as conscious dancers who deeply respect and honor the world around us.

Figure B| The figure shows a kinetic sculpture by the artist Daniel Wurtzel (Museu do Amanhã in Rio de Janeiro). The different flows on the planet are represented by pieces of fabric dancing.



Source: JRC

COMMENT

The High Seas Treaty: From Genetic Resources to Planetary Stewardship

Video 1| A visual celebration of Earth's interconnected systems highlighting the "four oceans" of light, air, water and land that sustain life in harmony. (Source: Museu do Amanhã in Rio de Janeiro). Watch the video at the following link: The "four oceans" of life - Museu do Amanhã - Rio de Janiero.



Source: JRC

COMMENT

Virus vs. Bacteria: AI-Designed Bacteriophages Combat Antibiotic Resistance

Giulia Panzarella

Antibiotic resistance remains one of the most urgent global public health threats, with the World Health Organisation (WHO) declaring it a critical priority due to its capacity to undermine effective treatment of infectious diseases [1]. Resistance naturally arises through bacterial evolution, allowing microbes to evade antimicrobial agents. This phenomenon has been accelerated by inappropriate antibiotic use in human healthcare and extensive application in agriculture and animal husbandry [2]. Bacterial resistance mechanisms involve enzymatic antibiotic degradation, modification of antibiotic binding targets, efflux pumps, and biofilm-mediated protection [3]. These mechanisms have evolved over millions of years, as antibiotics themselves are natural microbial metabolites used in microbial competition. Antibiotic selective pressure, especially in hospitals, drives the emergence and spread of resistant strains [4]. The scale of the AMR crisis is projected to escalate dramatically; by 2050, antimicrobial-resistant infections could cause nearly 10 million deaths per year globally, surpassing cancer fatalities [5].

The European Union (EU) has implemented comprehensive policies to mitigate AMR. Very recently it has been announced the launch of the European Partnership on One health Antimicrobial Resistance (OHAMR) [6], a €253 million initiative to tackle antimicrobial resistance through research and innovation. The EU also recognises the environment as a critical reservoir and transmission pathway for resistant bacteria and antibiotic residues, and recent regulatory updates have focused on wastewater management as a key intervention point. The recast Urban Wastewater Treatment Directive (UWWTD, Directive (EU) 2020/1161) seeks to reduce the release of pollutants, including antimicrobials and resistant bacteria, into water bodies by enforcing more stringent standards for urban wastewater treatment [7]. This legislative framework is integral to controlling environmental dissemination of antibiotics and resistance genes, complementing efforts to reduce overall antibiotic consumption in healthcare and agriculture [8].

Bacteriophages - viruses that specifically infect bacteria - offer a targeted alternative to conventional antibiotics [9]. Their application in treating resistant bacterial infections has been revisited with growing interest, especially as tailored phage cocktails can reduce the emergence of resistance. A transformative advancement is the integration of artificial intelligence (AI) with synthetic biology to engineer bacteriophages. A recent preprint study [10] documents the use of Evo 2, an advanced AI model trained on over 128,000 genomes, to design novel phage genomes. Starting from ΦX174, a well- characterised single-stranded DNA phage infecting *Escherichia coli*, Evo 2 generated thousands of candidate genomes; sixteen were experimentally verified to effectively kill multidrug-resistant *E. coli* strains in vitro (Figure C). This approach accelerates phage development timelines and offers customisation to emerging resistant bacterial threats, representing a powerful complement to existing strategies [11]. Complementing AI-driven phage engineering, recent studies demonstrate AI's capability to design antimicrobial proteins. An Australian group employed machine learning to create proteins that kill resistant *E. coli* strains, published in *Nature Communications* [12].

COMMENT

Virus vs. Bacteria: AI-Designed Bacteriophages Combat Antibiotic Resistance

Additionally, breakthroughs with AlphaFold, capable of predicting protein structures from sequence data with high accuracy [13] have revolutionised drug discovery workflows.

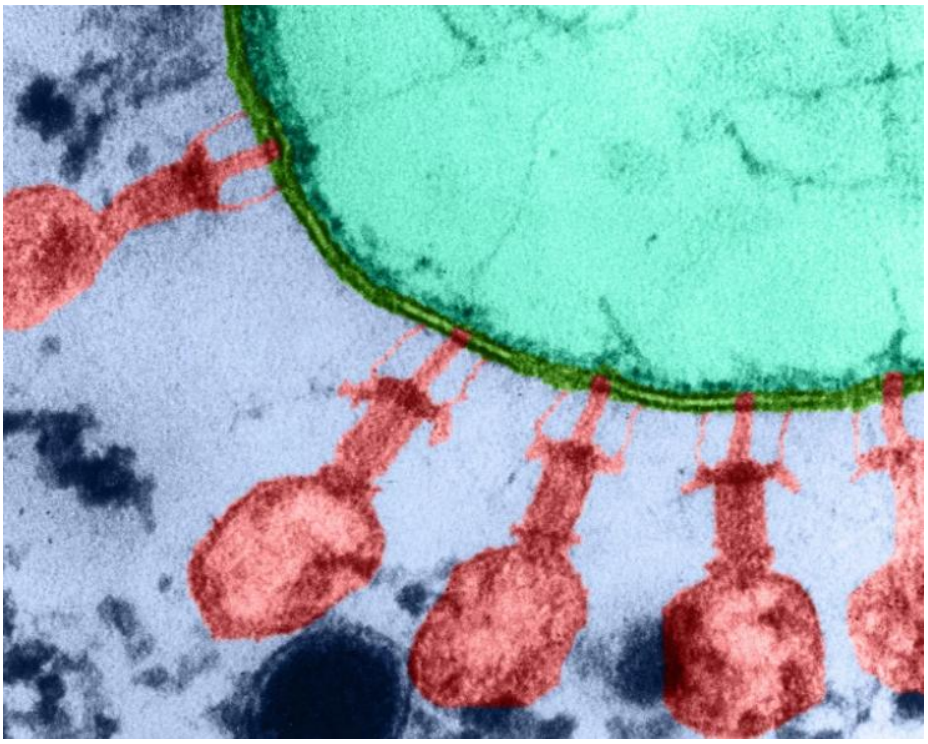
Collectively, these innovations highlight a new era of "augmented intelligence" in biomedical research, marrying human expertise with AI to overcome the limitations of traditional drug development.

Despite promising results, challenges remain for clinical translation. Phage therapy requires rigorous regulatory frameworks to ensure manufacturing quality, safety, efficacy, and to monitor the evolution of bacterial resistance to phages. AI-designed biological agents demand comprehensive validation to address potential off-target effects and immunogenicity.

From an environmental perspective, implementing and enforcing EU directives like the recast UWWTD is critical to minimising the environmental reservoirs of antibiotic residues and resistant organisms, thereby preventing further amplification and spread.

Future progress will depend on integrating multi-omics data, adaptive AI learning systems, and cross-sectoral policies aligned with the One Health framework to sustainably control AMR.

Figure C| AI-designed bacteriophages were capable of infecting and killing host bacteria..



Source: Lee D. Simon/Science Photo Library

EU SUPER-SITES: Insights from the Outsight

Sara Comero, Simona Tavazzi, Natalia Głowacka, Giulia Panzarella, IT Office

The EU SUPER-SITES form an integral component of GLOWACON (Global Consortium for Wastewater and Environmental Surveillance for Public Health), an ambitious global initiative launched in 2024 to create an international sentinel system for early detection and real-time monitoring of epidemic threats and emerging public health risks.

SUPER-SITES act as advanced sentinel outposts positioned at strategic locations across Europe, designed to provide high-resolution data on pathogens circulating among populations and travelers.

Their key role is to link community-based disease surveillance with environmental data, primarily through wastewater monitoring at critical transit hubs such as airports, border crossings, and densely populated urban centres.

As of September 2025, 21 active SUPER-SITES span 44 locations across the EU, collectively having collected 167 samples with 82 subjected to PCR analysis for pathogen detection. This network exemplifies a highly coordinated, cross-sectoral cooperation, advancing GLOWACON's vision of embedding wastewater surveillance as a routine, globally interoperable public health tool.

EU SUPER-SITES employ standardised protocols for sample collection, molecular diagnostic testing, and data integration; activities coordinated by the EC Joint Research Centre. Wastewater samples undergo quantitative PCR assays targeting a broad spectrum of viral, bacterial, and protozoan pathogens known to pose epidemic or pandemic risk.

These efforts will be complemented by metagenomic sequencing, integrated with biological (pathogens), chemical (toxins, illicit drugs), and bioinformatics analyses to track pathogen variants and emerging threats in near real time. This real-time environmental surveillance augments traditional clinical reporting, improving sensitivity and timeliness of outbreak detection, especially for pathogens with high rates of asymptomatic transmission or limited clinical testing access.

SUPER-SITES foster synergies with other surveillance systems including event-based community surveillance, thereby integrating diverse data streams into a holistic GLOWACON surveillance framework: GLOWACON Sentinel System. They represent a scalable model with potential expansion beyond Europe to foster international collaboration and harmonisation of wastewater-based epidemiology (WBE) standards.

Challenges ahead include harmonising data governance and privacy frameworks, addressing technical variability across sites, and integrating AI-driven analytics for predictive outbreak modelling. Strengthening capacity-building and fostering multi-disciplinary partnerships remain essential for maximising SUPER-SITES' potential.

Looking forward, SUPER-SITES aim to expand pathogen panels, incorporate antimicrobial resistance markers, integrate environmental and social determinants data, and promote the exchange of best practices. These enhancements will support a comprehensive, early-warning public health infrastructure aligned with the EU's strategic health security objectives.

GLOWACON NEWS

Wastewater is the new intelligence – Outcomes from EXPO 2025

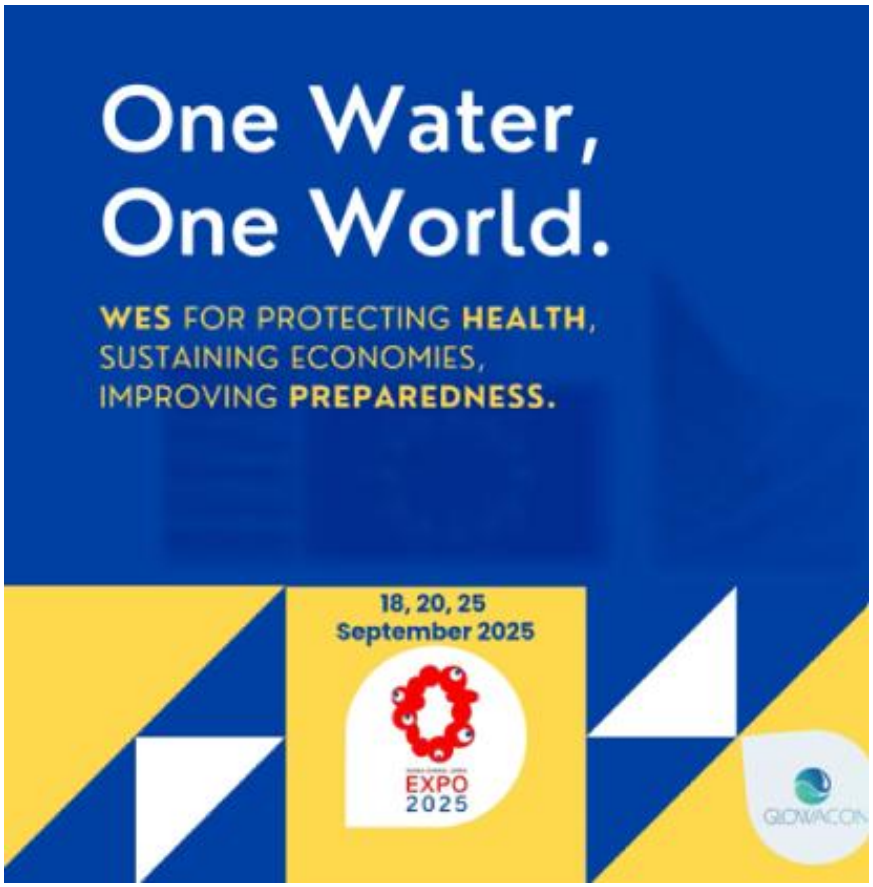
Bernd Manfred Gawlik, Angela Tessarolo, Bilge Alpaslan-Kocamemi

The EXPO OSAKA 2025 served as a crucial international platform for advancing discussions on innovative water and health solutions, directly aligning with the EXPO's theme of "Designing Future Society". During September 2025, high-level sessions, organised collaboratively by the European Commission's Joint Research Centre (JRC), the Health Emergency Preparedness and Response Authority (HERA), the Turkish Water Institute (SUEN), the World Water Council (WWC), and the Regional Government of Madeira, established a transformative vision for environmental surveillance and integrated water management (Video 2). The first key event "GLOWACON at the Expo Osaka," was organised jointly by HERA and the JRC at the EU DEL Pavilion.

The session focused on solidifying Wastewater and Environmental Surveillance (WES) as an innovative, scalable, and non-invasive system capable of detecting infectious threats early, often before clinical testing is sought.

High-level attendance included the Deputy-Head of DG HERA, the Mayor of Osaka (Hideyuki Yokoyama), the President of the World Water Council (Loïc Fauchon), and the Deputy Minister of Agriculture and Forestry of Türkiye (HE Ebubekir Gizligider).

Video 2| Wastewater is the new intelligence! Key takeaways video from EXPO OSAKA. The 'One Water, One Health' framework - EXPO OSAKA 2025.



Source: JRC

Key messages highlighted that WES must integrate sectors beyond traditional Public Health to succeed, also a necessity illustrated by the new European Urban Wastewater Treatment Directive (UWWTD).

Methodological discussions emphasised the shift toward metagenomics to enable near-infinite pathogen detection capacity, ensuring WES acts as an essential tool for pandemic preparedness in a globalised, federated approach. A major objective was identified: the need to break down historical silos between academic communities by bridging biological monitoring (pathogens, AMR) and chemical monitoring (illicit drugs, toxins) to enhance sensitivity, speed, and innovation.

GLOWACON NEWS

Wastewater is the new intelligence – Outcomes from EXPO 2025

The event saw the launch of GLOWACON Aviation, a new initiative prioritising WES at international airports as a critical frontier for pathogen monitoring.

In this regard a dedicated panel addressed WES at strategic locations to monitor cross-border health threats. Samples were collected at three Japanese airports: Haneda, Narita (major hubs), and Fukuoka (regional).

The potential for WES to support clinical settings (such as ICUs) was also discussed, despite challenges like high sample dilution due to therapies. The strong interest from the European Network of Intensive Care units, organized as part of the European Society of Anaesthesiology and Intensive Care Medicine (ESAIC), was highlighted.

The "Water, Health and Competitiveness" conference, organised by the Turkish Water Institute (SUEN), the World Water Council (WWC), and the JRC, defined the **'One Water, One Health'** framework in action.

The session advanced the 'One Water' approach, which recognizes all water (in cities, industries, or farms) as part of one integrated cycle, linking human, animal, and environmental health.

This holistic vision explicitly connects water, environment, food, and energy systems, forming the WEFE-H nexus.

The goal is to ensure water management not only sustains health but also maintains economic competitiveness and food security.

In this regard, Ebubekir Gizligider, Deputy Minister of Agriculture and Forestry of Türkiye, delivered a keynote, highlighting Türkiye's pioneering role in developing a comprehensive WES system that has expanded from SARS-CoV-2 surveillance to include antimicrobial resistance, food safety, and animal health.

A significant focus was also placed on transformative innovations, particularly the concept of *Ecofactories.*

These are modern hubs that move beyond traditional wastewater treatment by recovering energy, recycling water, and turning waste into valuable resources.

According to the President of the WWC, Loïc Fauchon, this new water sector vision, embracing the full character of WES as a practical One Health Tool, will "shape the discussion around the value of water".

The event concluded with key announcements, including the launch of the call for the EU Water Academy, aimed at strengthening global capacity building in advanced water systems. The event included a live art performance of Japanese Calligraphy. The final series of sessions, hosted at the Portuguese Pavilion, focused on island sustainability through the Islands-of-Life Initiative, organised by the Regional Government of Madeira Island and the JRC.

The Initiative was formally opened by Dr. Miguel Filipe Machado de Albuquerque, President of the Regional Government of Madeira Island. The 'Islands-of-Life Initiative' was launched to unite various global island groups (such as Macaronesia and the Caribbean) to collectively address shared vulnerabilities, including resource scarcity, climate resilience, and health security.

The event positioned Madeira Island as a global leader in innovative water and health solutions; the country was formally recognised as the first official EU Super Site Sentinel System member for wastewater surveillance and was also highlighted as a potential future training hub for the EU Water Academy.

GLOWACON NEWS

Wastewater is the new intelligence – Outcomes from **EXPO 2025**

September 25th was reserved for a private sector session focused on the "Business Angles to Wastewater-Encoded Data" and facilitating new Public-Private Partnerships (PPPs).

This meeting included match-making and speed-dating sessions designed to invigorate collaborations between public agencies and private sector entities. The goal was to identify synergies and explore joint actions necessary for forging win-win PPPs to scale WES solutions and 'Ecofactory' models. The discussions focused on specific use cases across communities of practice, including Public Sector, ICUs/hospitals, Security, Tourism, and Aviation. The aim of these interactions was to pave the way for partnerships that commit to dedicated and concrete actions of critical dimension supporting health and preparedness objectives.

The full set of events concluded with a ceremony and a cocktail reception featuring the OSAKA EXPO light show and a Calligraphy Performance by Tominga Haruto.

Figure D| Wastewater Treatment Group – Osaka..



Source: JRC

Figure E| Sample Collection Point – Osaka



Source: JRC

Figure F| Wastewater Treatment Plant – Osaka



Source: JRC

THE HUMAN SAMPLE
A Space for Societally Relevant Perspectives

Editorial Team

This section is designed to bridge the gap between technical data and real-world impact by featuring qualitative, societally relevant perspectives that illuminate the practical applications and benefits of wastewater surveillance initiatives. This section aims to capture qualitative and societally meaningful perspectives in the form of testimonials, reflections, and narratives that highlight how wastewater surveillance data translates into real-world impact. By focusing on lived experiences and practical insights, *The Human Sample* enriches the technical and scientific discussions with human-centered viewpoints, emphasising the societal value and relevance of surveillance data beyond laboratory results.



Source: JRC

Invitation to Contribute

We warmly invite experts, practitioners, community representatives, and other stakeholders to share their reflections, experiences, and stories related to wastewater surveillance and its role in public health, policy, and beyond. Contributions to T*he Human Sample* can take the form of short testimonials, opinion pieces, or narrative accounts that provide qualitative depth and societal context to the data-driven findings presented in this and future issues.

Submission Guidelines

Those interested in contributing are encouraged to contact the editorial team via email at JRC-EU4S-DEEP@ec.europa.eu. Please include "[BLUEMIRROR]" explicitly in the subject line of the email to facilitate processing of submissions.

We look forward to inaugurating a rich and engaging dialogue that bridges scientific evidence with human experience in eco-health surveillance.

Thank you for your cooperation and dedication.

BRIEF COMMUNICATIONS

THE EUROPEAN WASTEWATER SURVEILLANCE DASHBOARD

The EU Wastewater Observatory for Public Health has launched a new tool: the European Wastewater Surveillance Dashboard. This innovative platform tracks the spread of infectious diseases by analysing wastewater across Europe, providing real-time insights on pathogen trends and emerging health threats to support rapid responses to disease outbreaks.



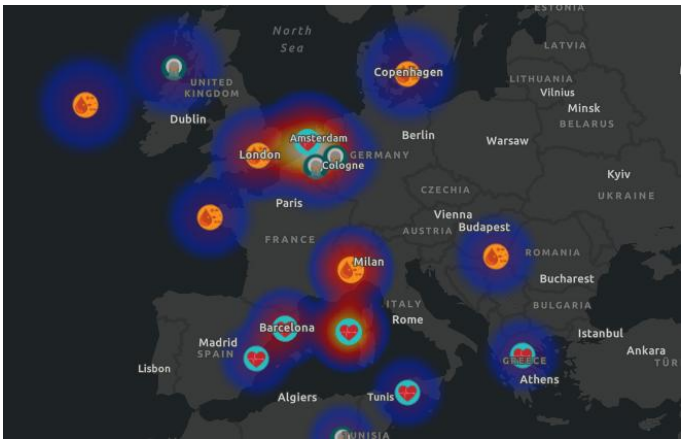
Try the tool and help us improve it - we invite you to explore the dashboard and send your feedback

ENCYCLOPAEDIA CLOACAE

Stay up-to-date with the latest information on pathogens and public health indicators in wastewater through the EU Wastewater Observatory's Encyclopaedia Cloacae.
This valuable resource supports disease detection and public health efforts across Europe.

Get involved and contribute to the growth of the Encyclopaedia by submitting publications or your organisation's logo to become an official contributor. Our team will review and add your contributions to the platform. To contribute, please visit the "Add Your Knowledge" area.

Advanced Heatmap Dashboard for Wastewater Pathogen Research Monitoring



Explore global research on pathogens in human excreta and wastewater through an interactive dashboard, featuring a dynamic heatmap of active research areas.

BRIEF COMMUNICATIONS



Are you ready to contribute to Europe's health preparedness? The European Commission is inviting expressions of interest from locations committed to using wastewater surveillance as a tool for early detection of health threats. If your site can support regular sampling—especially near transport hubs—and aligns with national or local surveillance efforts, you could become part of the European Super-Sites network. The Commission covers all sample dispatch and analysis costs and supports partnerships with key sectors.

Submit your interest now via the Super-Site call for expression of interest:
👉https://ec.europa.eu/eusurvey/runner/EU4S_SuperSites

GLOWACON

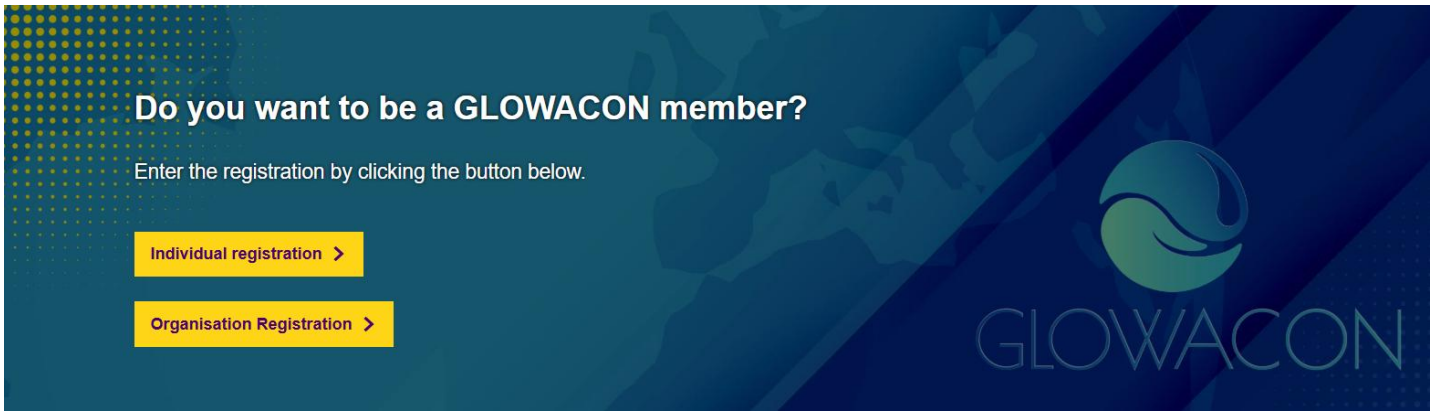
GLOWACON Aviation Wastewater Surveillance Programme



With the increasing interconnectivity of global travel, the aviation sector presents a significant focus for innovative public health surveillance methods. This initiative assembles public health services, airlines, international organizations, airport authorities, academic institutions, and contracted service providers to enhance pathogen monitoring through wastewater analysis.

Through the GLOWACON Aviation Programme, we aim to address the current limitations in public health surveillance by working with the aviation sector, demonstrating the practicality and effectiveness of wastewater-based epidemiology. The insights gained will advance WES/WBE as a standard component of international public health efforts, ensuring rapid pathogen detection in a globally mobile population. Participants will not only contribute to global health safety but also benefit from cutting-edge research and data analysis opportunities, while staying in control of the derived data. Join the future of Wastewater Surveillance.

BRIEF COMMUNICATIONS

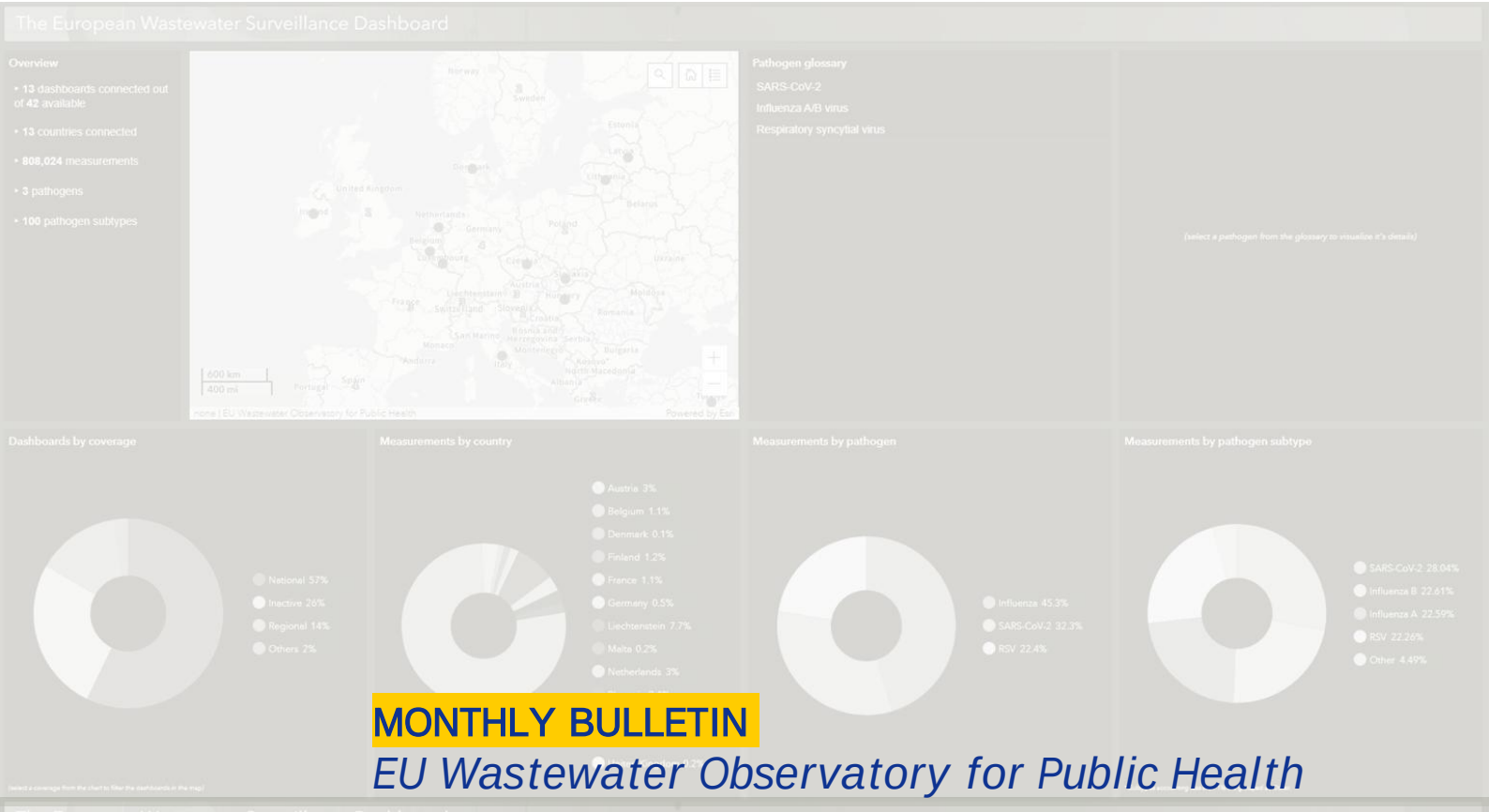


Individual registration
Organisation Registration



A flagship initiative to strengthen water resilience in Europe The European Union is grappling with substantial water challenges due to climate change, unsustainable practices, ageing infrastructure, and mismanagement.
In response, the European Commission has introduced the European Water Resilience Strategy, with the water efficiency first principle, featuring EUWA, the European Water Academy, as one of its flagship initiatives.
Stay tuned for further announcements.

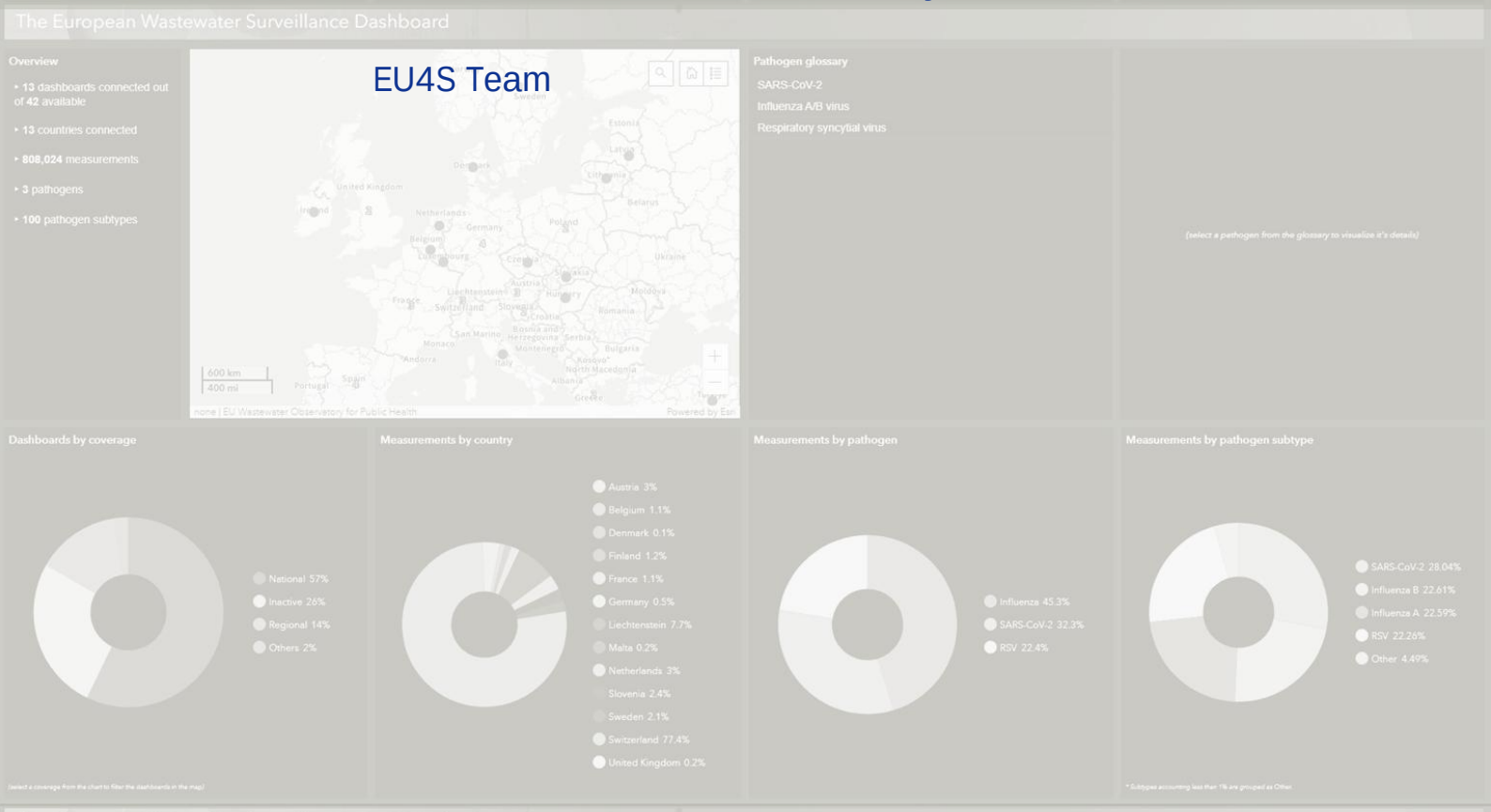
Check out the EUWA Fact Sheet.



MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

EU4S Team





MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

EU4S Team

EXECUTIVE SUMMARY

- Updated SARS-CoV-2 wastewater data are available at the national level for 17 of the 27 EU countries.
- As of September 2025, 5 countries (Austria, Denmark, Germany, Hungary, Latvia) have reported increasing or slightly increasing trends in SARS-CoV-2 levels in wastewater. Greece has reported stable levels, while Finland and Netherland have reported decreasing levels. Viral loads are classified as very low in Greece and moderate in 3 countries (Belgium, Denmark, Luxembourg).
- In September 2025, the XFG variant (Stratus) became the most dominant lineage of SARS-CoV-2 in Europe, quickly surpassing other Omicron subvariants. At this time, there is no evidence to suggest that XFG subvariants are more likely to cause severe illness compared to previous Omicron subvariants.
- Influenza and RSV continue to be monitored in Austria, Belgium, Finland, Germany, Hungary, Luxembourg and Sweden. Belgium, Finland, Ireland and Slovenia are also conducting tests for poliovirus but currently no new data have been reported. Sweden and Luxembourg conducted surveillance for enteric viruses until mid-2024.
- Dengue, Chikungunya, Zika, and yellow fever are emerging as significant arboviral diseases of growing concern. However, at present, only sporadic wastewater surveillance data are available.
- The EU Wastewater Observatory for Public Health is keenly following these developments and will integrate new insights in forthcoming bulletins, with input from the EU Super-Sites.

This bulletin offers insights derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Additionally, data from existing regional/local, and non-EU sources are included. The most recent retrieval was performed on 26th September 2025.

Wastewater surveillance analyses samples to reveal community health and environmental trends, enhancing traditional public health monitoring; the EU Wastewater Observatory collects and shares this data to support decision-making and pandemic preparedness, tracking viruses like SARS-CoV-2, avian flu, mpox, RSV, and influenza.

In June 2025, the European Commission's Health Emergency Preparedness and Response Authority (HERA), in partnership with the Joint Research Centre (JRC), initiated the sample collection phase of the European Super-Sites initiative, implemented through a HaDEA (European Health and Digital Executive Agency) framework contract.

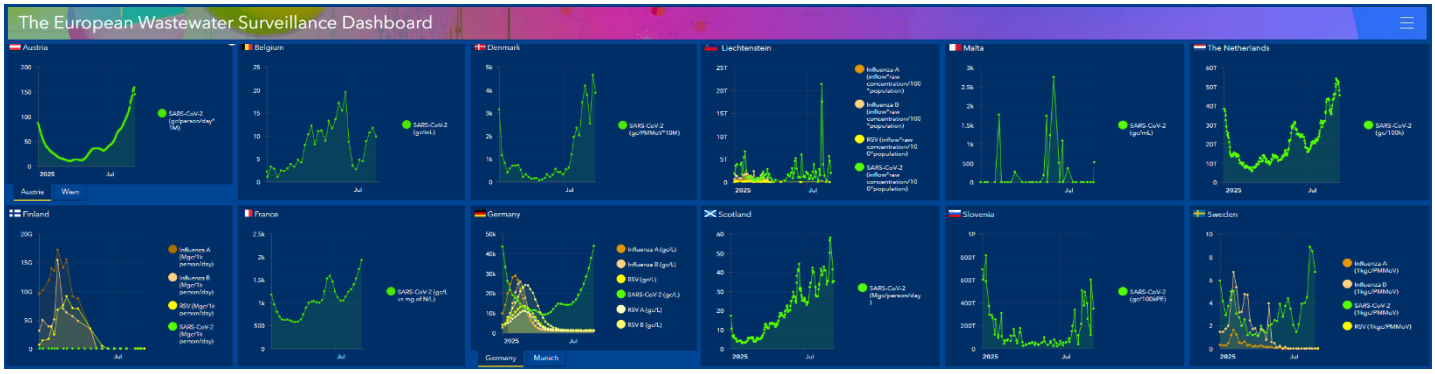
MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

This network of strategically located sites monitors wastewater near transport hubs to detect emerging health threats. As part of the Global Wastewater Sentinel System, the initiative enhances early warning capabilities and supports public health decision-making across the EU and beyond.

SARS-CoV-2 surveillance

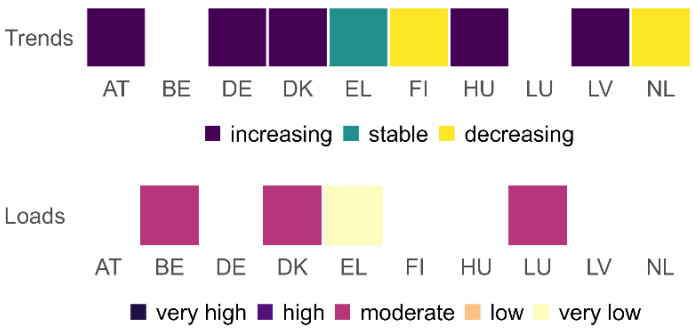
Figure 1. Overview of the European Wastewater Surveillance Dashboard: near real-time wastewater data on measured pathogens for connected countries, providing early warnings and rapid responses to protect public health across Europe.



Source: The European Wastewater Surveillance Dashboard.

A summary of SARS-CoV-2 trends and viral loads is provided (Figure 2) for various EU countries. Figure 2 includes only those countries for which data on viral loads and trends is available. If information is missing, it indicates that either the national trend or viral load has not been specified, or the data has not been updated to reflect the current month.

Figure 2. Viral loads and trends of SARS-CoV-2 across EU countries.



Source: EC-JRC

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

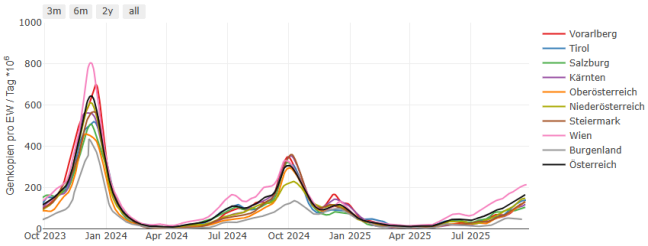
AUSTRIA

Last data update: 23 September 2025

Austria's monitoring of SARS-CoV-2 involves 48 strategically chosen sewage treatment plants, encompassing over 58% of the country's population.

SARS-CoV-2: In week 39, a slight increase in viral loads has been observed, keeping the trend indicator above 1. In week 37, the variant XFG was the most prevalent, including its sub-lineages, which accounted for approximately 65% of reported cases.

Figure 3. Person-weighted trends in the federal states (and Austria as a whole).



Source: Abwassermonitoring.

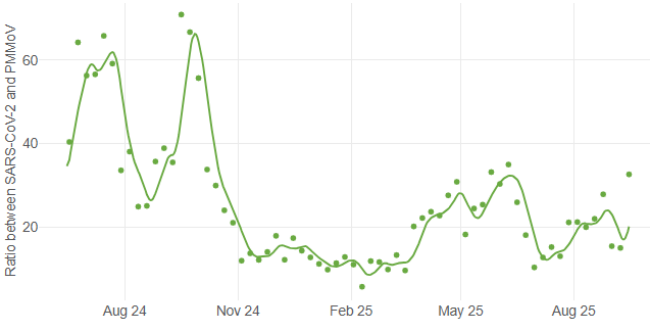
BELGIUM

Last data update: 15 September 2025

The surveillance program encompasses 38% of the Belgian population, involving the collection of wastewater samples once a week from influents of 30 wastewater treatment plants distributed across Belgium, mainly in areas of high population density.

SARS-CoV-2: In week 38, the national level of SARS-CoV-2 was classified as moderate. In August 2025, the XFG variant was the dominant strain.

Figure 4. SARS-CoV-2 viral to faecal concentration at a national level.



Source: Wastewater surveillance Respiratory viruses Sciensano Belgium

CZECH REPUBLIC

Last data update: 19 August 2025

Systematic monitoring of wastewater for the presence of SARS-CoV-2 has been conducted since February 2023 at five wastewater treatment plants connected to the largest cities in the Czech Republic and Václav Havel Airport in Ruzyna. The monitored collection points cover approximately 21% of the Czech Republic's population.

SARS-CoV-2: The dashboard, updated to week 34, shows SARS-CoV-2 loads in each treatment plant separately, while at the national level, the presence of the viral load is given as a percentage of positive detections. Currently, this information is only available in Czech.

MONTHLY BULLETIN

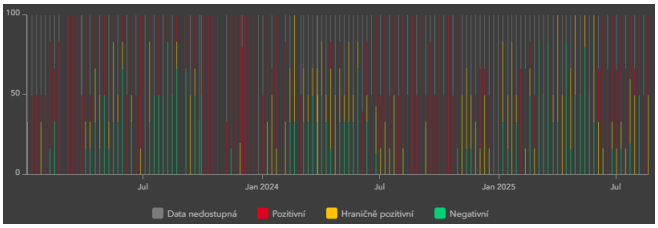
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

CZECH REPUBLIC

Last data update: 19 August 2025

Figure 5. Presence of SARS-CoV-2 viral load in monitored wastewater.



Source: Coronavirus wastewater monitoring weekly report

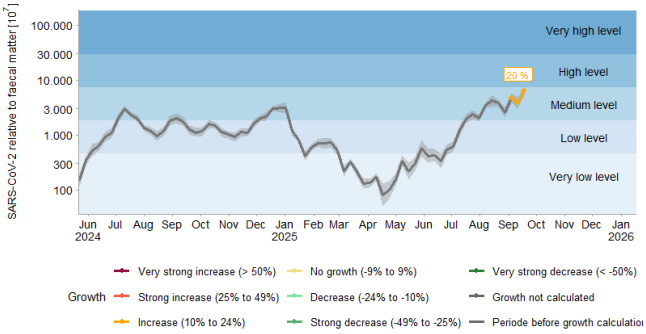
DENMARK

Last data update: 21 September 2025

Denmark's national surveillance of SARS-CoV-2 in wastewater covers 49% of the population. Twice a week, samples are collected from 29 wastewater treatment plants. The results from the wastewater are compiled for the entire country, as well as for the five regions and each specific sampling location.

SARS-CoV-2: In week 38, the concentration of SARS-CoV-2 in wastewater across the country was at a medium level. The national growth rate, reflecting the average weekly growth over the past three weeks, indicates an increase in the SARS-CoV-2 wastewater concentration in the range of 10% to 24%.

Figure 6. Concentration of SARS-CoV-2 in the wastewater, aggregate for all the sampling sites in Denmark.



Source: National wastewater surveillance of SARS-CoV-2.

FINLAND

Last data update: 22 September 2025

Various wastewater treatment plants across different regions of Finland actively contribute to the monitoring efforts. Presently, about 44% of the wastewater generated by the Finnish population is included in the wastewater monitoring initiative.

SARS-CoV-2: The results of monitoring coronavirus in wastewater are available at both local and national levels, updated until week 39. The method of determination changed at the end of 2024. A statistical model estimates the trends, which reveals a decreasing trend in recent weeks.

MONTHLY BULLETIN

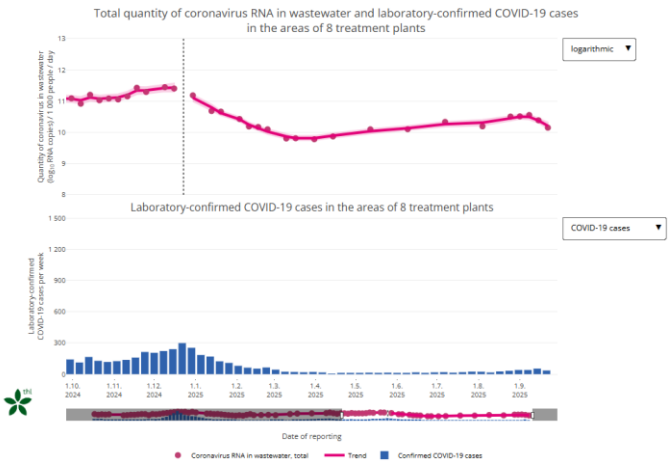
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

FINLAND

Last data update: 22 September 2025

Figure 7. Total quantity of coronavirus in wastewater and laboratory-confirmed COVID-19 cases, in the areas of 8 treatment plants.



Source: Coronavirus wastewater monitoring weekly report.

FRANCE

Last data update: 14 September 2025

The national wastewater monitoring system, SUM'Eau, was launched by the General Directorate of Health and the Directorate for Water and Biodiversity, with expert support from health agencies. The data allows for estimating the SARS-CoV-2 circulation trend in 54 treatment plants across 12 administrative regions.

SARS-CoV-2: The data has been updated to week 37 and is accessible in the form of CSV files. The SARS-CoV-2 monitoring indicators in wastewater are computed as the ratio between the viral concentration of SARS-CoV-2 and the concentration of ammoniacal nitrogen. The overall national indicator is also available.

Figure 8. Structure of the CSV file available containing Sars-Cov-2 wastewater surveillance of data.

Dernière mise à jour de l'exploration le 24/09/2025 à 18:00 - *Format csv*(106 KB)

sumeau-indicateurs-2025-09-24-18h00.csv

Surveillance du Sars-Cov-2 dans les eaux usées (SUM'Eau) - *Santé publique France*

Données | Description | Structure de données | Métadonnées

semaine	AMIENS	ANGERS LA BAUMETTE	ANGLET (PONT DE L'AVEUGLE No2)
2025-S37	5729.57857906634	4228.43695900661	614.485442822214
2025-S36	4279.72689411237	3055.74331862147	625.018648708894
2025-S35	3412.37619498177	2377.94956028132	689.284869403674

Source: Data Explorer data.gouv.fr.

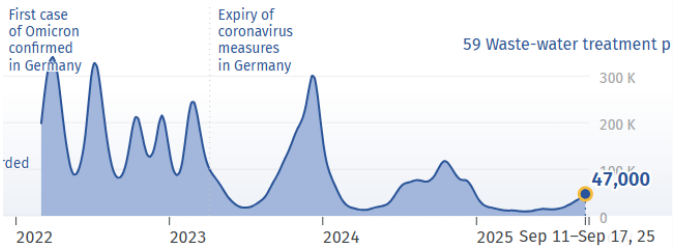
GERMANY

Last data update: 17 September 2025

The viral load of SARS-CoV-2 in wastewater is currently assessed at 169 wastewater treatment plants, providing a coverage for approximately 32% of the population. The number of reporting locations fluctuates over time.

SARS-CoV-2: During the past 4 weeks (reporting week 38), the average SARS-CoV-2 viral load 59 reporting wastewater treatment plants increased.

Figure 9. Mean viral load in wastewater.



Source: Infection Radar.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

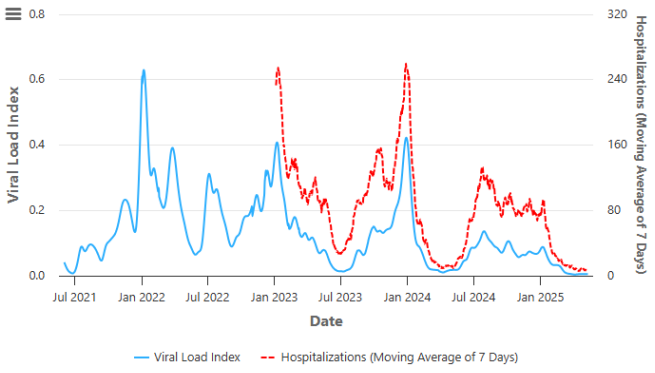
SARS-CoV-2 surveillance

GREECE

Last data update: 21 September 2025

The dashboard presents data from the National Wastewater Epidemiology Network (NWEN), which monitors SARS-CoV-2 and other respiratory viruses in urban wastewater across 10 regions in Greece, covering 70% of the population. It visualises weekly viral load trends at both city and national levels using maps and charts, based on frequent sampling and analysis by coordinated laboratories. The dashboard also includes data on pharmaceuticals and chemicals found in wastewater, offering a comprehensive tool for public health surveillance and early warning.
SARS-CoV-2: During week 38, the national viral load was very low, showing a -0.8% change compared to the previous week, which indicates a stable trend. Among the 8 monitored Wastewater Treatment Plants, 2 displayed an increasing trend, 5 showed a decreasing trend, and 1 remained stable.

Figure 10. National index variation in viral load levels.



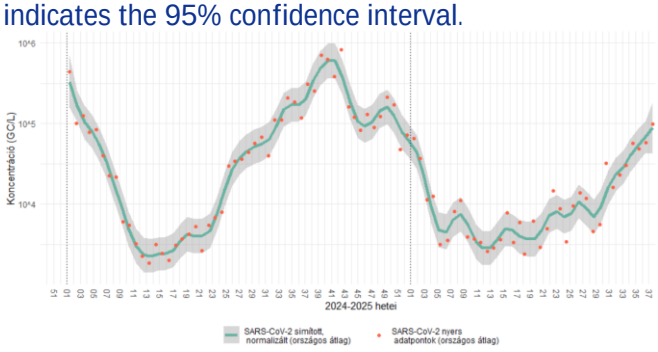
Source: Wastewater surveillance dashboard.

HUNGARY

Last data update: 14 September 2025

The data for wastewater monitoring in Hungary are obtained from three wastewater plants serving the entire area of Budapest and a portion of the agglomeration. Additionally, 19 wastewaters from five other agglomeration settlements are examined as a combined sample. Collectively, these analyses encompass over 40% of the domestic population.
The weekly report is currently only available in Hungarian.
SARS-CoV-2: During week 37, the national average concentration of SARS-CoV-2 in wastewater showed a slight increase. Specifically, six areas experienced an upward trend, twelve areas displayed stagnation, and a decrease was noted in one region. By week 50 of 2024, the newly emerged XEC subvariant had become the dominant strain.

Figure 11. National, population-weighted average of SARS-CoV-2 copy number measured in wastewater. The red dots show the raw data, the grey band indicates the 95% confidence interval.



Source: National Center for Public Health and Pharmacy - Weekly wastewater inspection results.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

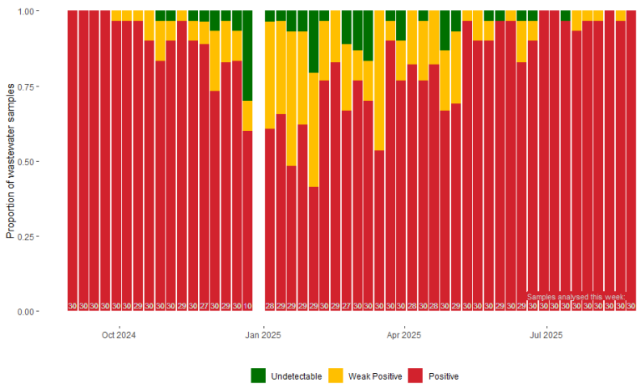
IRELAND

Last data update: 30 August 2025

Since January 2023, Ireland's National Wastewater Surveillance Programme has been conducting weekly sampling in 30 wastewater catchment areas. The 30 catchment areas, covering 70% of the population connected to public wastewater treatment facilities, are distributed across all regions of Ireland.

SARS-CoV-2: In week 35, samples were collected and analysed from 29 out of 30 catchment areas. SARS-CoV-2 RNA was detected in 100% of these samples. Among the 29 catchment areas where comparisons with the previous week were possible, the viral load remained stable (within +/- 10%) in 19 areas. One area experienced a decrease of 10% or more, while 9 areas recorded an increase in viral load ranging from 10% to 50%.

Figure 12. Wastewater SARS-CoV-2 detections in Ireland. Week 26 2025 (22/06/2025 to 28/06/2025).



Source: Report week 35.

ITALY

Last data update: 06 December 2024

Wastewater monitoring in Italy is conducted through Flash Surveys, comprising monthly sampling campaigns at various locations across the country over a short duration.

SARS-CoV-2: The monitoring results are usually reported every two months, with an average delay of a few months from sample collection to reporting. The latest update is from December 2024, where a total of 111 wastewater samples were collected from 16 regions and 2 autonomous provinces (A.P.). Mutations characteristic of the Omicron variant were identified in 9 regions and A.P., including those associated with the XEC*, KP.3*, LF.7*, and LP.8.1 sublineages.

LATVIA

Last data update: 19 September 2025

The national monitoring program in Latvia focuses on identifying the presence of SARS-CoV-2 in wastewater across municipalities. Twice a week, untreated wastewater samples are gathered from various facilities within municipal centralised wastewater collection systems. The weekly report is currently only available in Latvian.

SARS-CoV-2: In week 38 the overall trend of RNA concentration in Latvia was increasing. Detailed reports on the prevalent variants are available for each wastewater facility, revealing that Omicron KP.3 sub-lineage was the most common variant in most locations during week 37.

MONTHLY BULLETIN

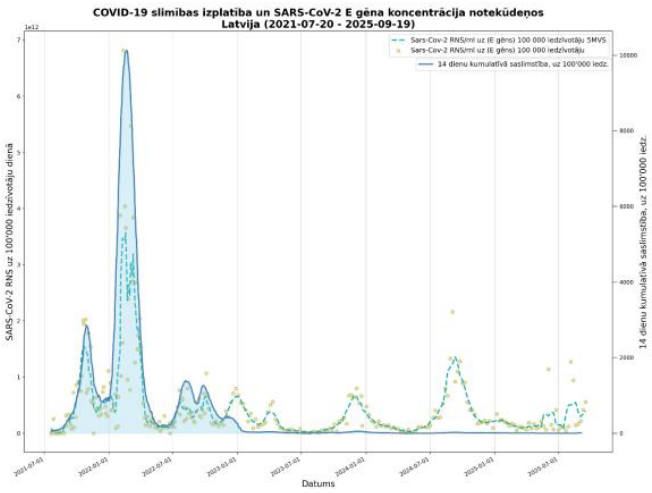
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LATVIA

Last data update: 19 September 2025

Figure 13. Spread of COVID-19 and detection of SARS-CoV-2 in wastewater:



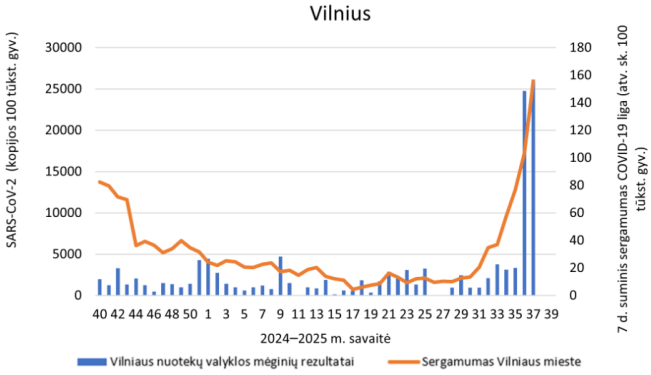
Source: Wastewater monitoring – Latvia.

LITHUANIA

Last data update: 14 September 2025

Lithuania has implemented a wastewater surveillance system to detect the SARS-CoV-2 virus and its variants since November 2022. Wastewater samples are collected weekly from three treatment plants.
SARS-CoV-2: Three wastewater treatment facilities are currently under surveillance. The website offers data for each municipality individually, and the latest updated data is from week 37.

Figure 14. Vilnius City Wastewater Treatment Plant (copies/ 100 000 pop.)



Source: Monitoring results 2024-2025 - National public health centre under the Ministry of Health.

LUXEMBOURG

Last data update: 15 September 2025

The program for monitoring the evolution of COVID-19 in wastewater in Luxembourg involves the collection of samples from eight strategically located wastewater treatment plants, covering five cantons and approximately 67% of the country's population. Results are published in fortnightly reports.
SARS-CoV-2: SARS-CoV-2 concentrations are reported at both the national and individual plant levels. In week 38, the national levels of SARS-CoV-2 are moderate.

MONTHLY BULLETIN

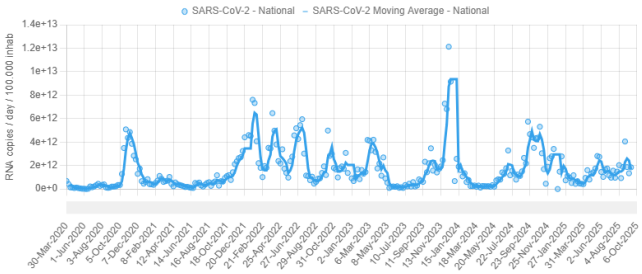
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LUXEMBOURG

Last data update: 15 September 2025

Figure 15. SARS-CoV-2 national levels (RNA copies/day/100.000 inhab).



Source: SARS-CoV-2 – MicrObs.

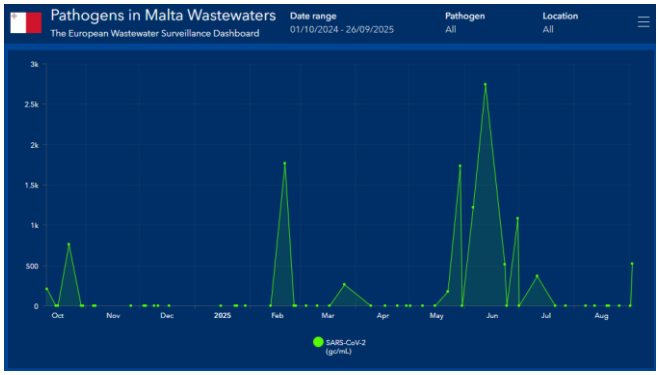
MALTA

Last data update: 02 September 2025

Malta does not currently have a dedicated national dashboard for monitoring SARS-CoV-2 levels in wastewater. However, the Ministry for Health and Active Ageing in Malta regularly submits updates on these measurements to the EU Wastewater Observatory for Public Health. Samples are collected from three wastewater treatment plants.

SARS-CoV-2: SARS-CoV-2 concentrations are reported at both national and individual plant levels. The latest results are updated to week 36.

Figure 16. Malta Dashboard.



Source: Pathogens in Malta wastewaters.

NETHERLANDS

Last data update: 21 September 2025

Wastewater samples are gathered at wastewater treatment plants throughout the Netherlands, ensuring comprehensive coverage.

SARS-CoV-2: During week 38, the levels of SARS-CoV-2 in wastewater showed a slight decrease in the national average, compared to the previous week. Multiple sub-variants of JN.1 are currently being detected in wastewater, particularly NB.1, XFG, LP.8.1.8.1, as well as sub-variants related to these three strains.

MONTHLY BULLETIN

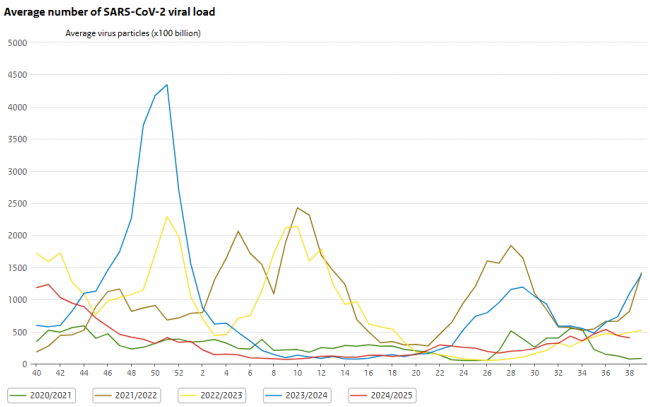
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

NETHERLANDS

Last data update: 21 September 2025

Figure 17. Netherlands. Average virus load (x100 billion) over time per 100,000 inhabitants.



Source: Weekly coronavirus SARS-CoV-2 figures | RIVM.

POLAND

Last data update: 29 January 2025

The dashboard is supported by a network of voluntary laboratories initiated by the National Institute of Public Health – National Institute of Hygiene (NIZP PZH - PIB). This network includes Provincial Sanitary and Epidemiological Stations, as well as local water supply and sewage companies. The SARS-CoV-2 wastewater data are normalised using PMMoV and reported for four cities, encompassing a total of 15 WWTPs.

SARS-CoV-2: Aggregated city data and specific site data are reported in plots. SARS-CoV-2 mutations detected in wastewater samples are reported for each city. The latest update is from the end of January 2025, and no further updates have been provided since the last monthly bulletins.

SLOVAKIA

Last data update: 16 March 2025

Wastewater surveillance for the presence of SARS-CoV-2 is conducted on a weekly basis in each region of Slovakia, with over 55 wastewater treatment plants being part of the project. Throughout the year, the monitoring intensity is adjusted based on the prevailing epidemiological situation.

SARS-CoV-2: The latest update is from March 2025, and no further updates have been provided since the last monthly bulletin.

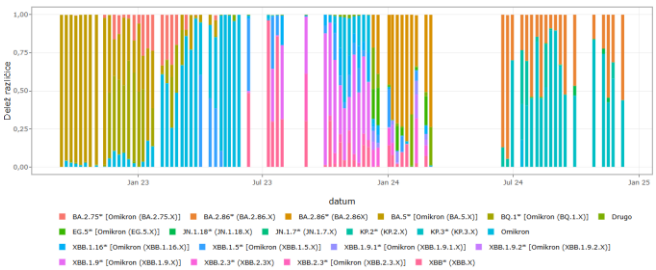
SLOVENIA

Last data update: 25 August 2025

From autumn 2022 and throughout 2024, Slovenia has been consistently monitoring SARS-CoV-2 in wastewater, using samples from 16 waste treatment plants across the nation. This monitoring network reaches about 34% of the Slovenian population and involves weekly sampling.

SARS-CoV-2: The dashboard displays SARS-CoV-2 wastewater monitoring data, including the percentage of circulating virus variants, for each individual wastewater treatment plant. The data has been updated to week 35, and results can be accessed via the Slovenian open data portal.

Figure 18. Mutation rates of the SARS-CoV-2 genome.



Source: C19-water.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

SPAIN

Last data update: 22 June 2024

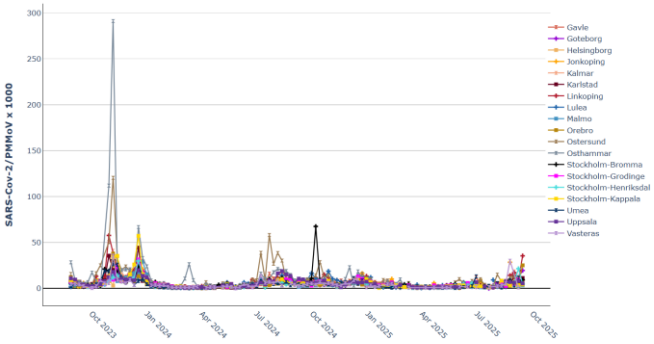
Wastewater surveillance in Spain involves the examination of the presence and evolution of the genetic material of SARS-CoV-2 RNA and the predominant variants in all treatment plants, totaling 54 facilities distributed throughout the country.
SARS-CoV-2: The summary report and dashboard have not been updated since week 25 of 2024.

SWEDEN

Last data update: 15 September 2025

The dashboard assesses SARS-CoV-2 levels in wastewater across various Swedish cities. For weekly SARS-CoV-2 monitoring, Sweden's public health agency collaborates with SEEC-SLU (the Swedish University of Agricultural Sciences), providing data covering 43% of the Swedish population with the broadest geographic reach across the country.
SARS-CoV-2: The levels of SARS-CoV-2 in wastewater are visualised using interactive plots.
The SEEC-SLU project provides weekly data updated to week 38, which can also be downloaded as a dataset.
The summary report, which is only available in Swedish, is currently not up to date.

Figure 19. PMMoV normalised levels of SARS-CoV-2 virus in multiple cities across Sweden.



Source: SARS-CoV-2 quantification | The Swedish Pathogens Portal.

REGIONAL EU DATA

The EU member states listed below furnish local or regional data regarding the surveillance of SARS-CoV-2 in wastewater for specific areas within their respective countries. The accompanying list includes accessible databases, along with their latest updates.

Austria

Vienna

Weekly wastewater analysis from the Vienna sewage treatment plant is part of the 'CSI Wastewater' project. The graphic shows the weekly occurrence of SARS-CoV-2. Data is updated to week 38, 2025.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

REGIONAL EU DATA

Germany

Berlin

The dashboard presents wastewater monitoring data from 4 treatment plants in Berlin. It shows SARS-CoV-2 data results in gene copies per litre of wastewater sampled from each plant. The analysis also includes circulating virus variants in Berlin, presented as a percentage of the total load. In week 36, the trends indicate a decrease in virus levels in three plants and an increase in one plant.

Bavaria

The dashboard shows wastewater monitoring data from the German state of Bavaria. It includes the SARS-CoV-2 trend, viral loads and variants in wastewater from thirty municipalities within Bavaria. In week 39, measurements were conducted at 21 locations. Among these, 94% showed an increasing trend, and 6% showed a stable trend.

Poland

The website provides surveillance data on the SARS-CoV-2 virus detected in sewage within the Warsaw agglomeration. The findings are presented through graphs that illustrate the normalised number of viral copies at four specific locations within the sewage network. The data is updated weekly, with results reflecting data up to week 38.

At the end of July and the beginning of August, there was a noticeable decrease in virus levels in samples collected from all locations.
However, in the following days, this trend reversed, and an increase in virus levels was recorded at all sampling points. This pattern continued throughout August, with the exception of the transition from August to September and was also observed in the subsequent weeks of September.

Spain

Madrid

The website presents visualisations of SARS-CoV-2 data in wastewater for the Community of Madrid. A map displays the progression of SARS-CoV-2 levels in wastewater across various municipalities and districts, but it is not updated monthly; the last update was in April 2025. Reports are available in Spanish.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

EU NEIGHBOURING COUNTRIES

This section provides wastewater monitoring details from neighbouring countries of the EU participating as observers in the EU National Contact Points (NCP) Mechanism. Links to national or local dashboards/websites and information about the latest updates are included.

Türkiye

Last data update: 13 July 2025

This website provides information on wastewater monitoring data in Türkiye, explicitly reporting SARS-CoV-2 viral load results. PDF reports are published for various regions (23 pilot cities), while Istanbul has its dedicated report. Wastewater samples are taken weekly from the city of Istanbul and bi-weekly from the other 23 cities, providing a coverage for approximately 41% of the population.
SARS-CoV-2: During week 28, the viral loads of SARS-CoV-2 were analysed in four wastewater treatment plants in Istanbul. The results indicated that no viral loads were detected in any of the plants. At the national level, 20 cities were tested in July (week 28), and all reported no cases across the wastewater treatment facilities examined.

Figure 20. SARS-CoV-2 viral load distribution map, Istanbul city



Source: COVID-19 Türkiye.

OTHERS

This section contains details on wastewater monitoring from non-EU countries. It includes links to national or local dashboards/websites and the latest updates, organised by continent.

EUROPE

Scotland

Last data update: 18 September 2025

The program has been operating in Scotland since June 2020. Each week, Scottish Water collects approximately 150-200 sewage samples from 115 wastewater treatment facilities across the country. These samples are then sent to NHS Lothian for testing (carried out by the Scottish Environmental Protection Agency until end July 2024).
SARS-CoV-2: Data is available at the national level, as well as for NHS Health Boards and local authority areas. The dashboard is updated to week 38.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

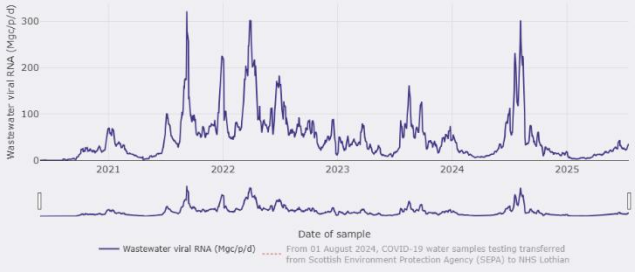
OTHERS

EUROPE

Scotland

Last data update: 18 September 2025

Figure 21. Seven-day average national trend in COVID-19 wastewater. Source: COVID-19 & Respiratory Surveillance.



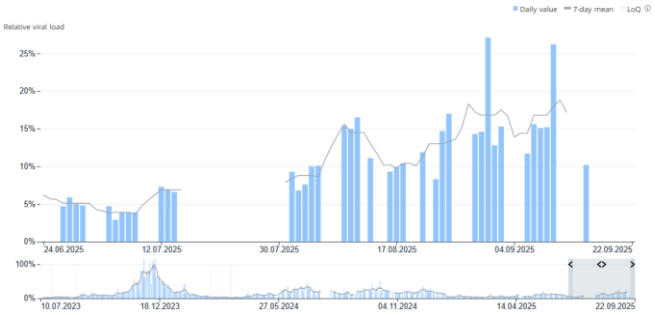
Switzerland

Last data update: 15 September 2025

The information published in the dashboard is based on data from Switzerland's national wastewater monitoring program, which has been tracking the viral load of SARS-CoV-2 since January 2022. As of January 2025, wastewater monitoring has been streamlined to include only 10 WWTPs in Switzerland, covering almost a quarter of the Swiss resident population, and one WWTP in the Principality of Liechtenstein. Additionally, variant monitoring is carried out at six WWTPs from December 2024.
SARS-CoV-2: Each region reports both the relative and absolute viral loads, as well as trends and variants.

During week 38, it was observed that viral loads remained stable in 9 regions but increased in 2 regions. The most frequently identified SARS-CoV-2 variant (latest update on 23.08.2025) was XFG in all the regions.

Figure 22. Relative viral load of SARS-CoV-2 at a selected location (Lugano).



Source: IDD - Infectious Diseases Dashboard.

AFRICA

South Africa

Last data update: 14 September 2025

The National Institute for Communicable Diseases (NICD) Centre for Vaccines and Immunology provides updated information on SARS-CoV-2 levels in wastewater across 14 districts in South Africa.
SARS-CoV-2: The dashboard presents SARS-CoV-2 levels both nationally and by district. Data is updated up to week 37. As of the beginning of June 2025, the most prevalent sub variant was N.B.1.8.X at 81%.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

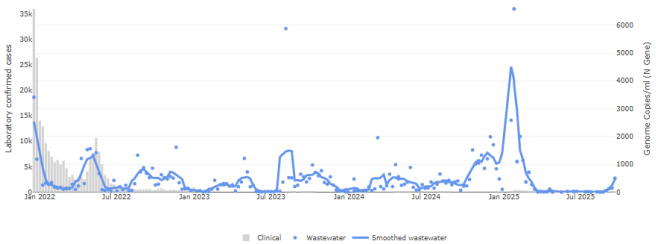
OTHERS

AFRICA

South Africa

Last data update: 14 September 2025

Figure 23. National SARS-CoV-2 wastewater levels.



Source: https://wastewater.nicd.ac.za/.

ASIA

India

Last data update: 27 July 2025

The dashboard displays the concentration of SARS-CoV-2 RNA copies per litre of wastewater in five major cities of Gujarat, India. Samples are collected from 40 wastewater treatment plants. The Lineage Prevalence section is also available.

SARS-CoV-2: The latest update on SARS-CoV-2 concentration in wastewater is from July 2025 (week 30). However, information on variants has not been updated since July 2024.

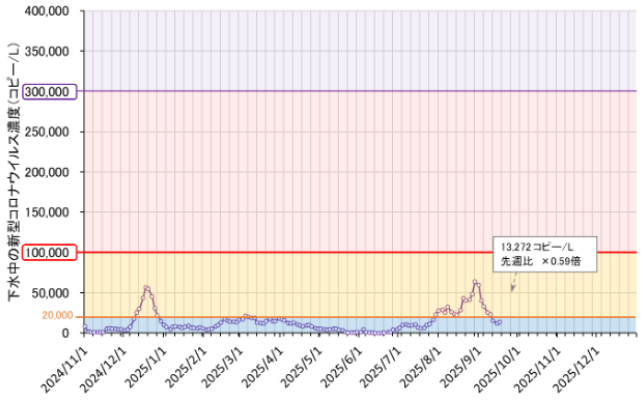
Japan

Last data update: 19 September 2025

The information presented on this website reveals the outcomes of regular assessments of COVID concentration in the wastewater of Komatsu City.

SARS-CoV-2: During week 38, SARS-CoV-2 levels in wastewater were low, showing a decreasing trend (0.59 times) compared to the previous week.

Figure 24. Virus concentration in Komatsu City expressed in copies/liter.



Source: Sewage Monitoring/Komatsu City Website.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

OCEANIA

Australia

Western Australia

Last data update: 19 September 2025

The dashboard presents an analysis of SARS-CoV-2 levels in the Perth Metropolitan area of Western Australia, starting from July 2022. As of August 2024, wastewater samples are collected weekly from three metropolitan wastewater treatment plants that serve approximately 79% of the city's population.
The concentrations of the virus genome in wastewater are shown as a 14-day moving average and have been updated to week 38. Omicron sub-lineage NB.1.8.X predominated across all three metropolitan wastewater sites (47.2%), followed by BA.3.2.X (20%).

New South Wales Australia

Last data update: 20 September 2025

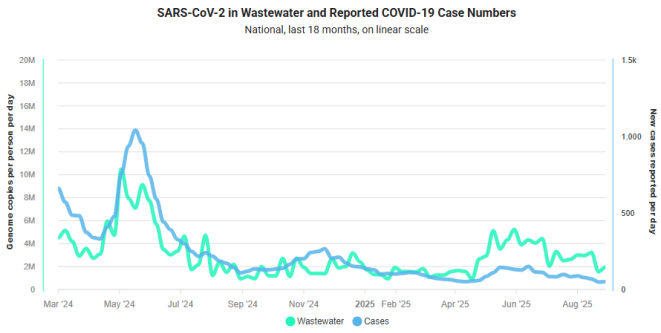
The New South Wales Sewage Surveillance Program collects samples from 4 sites in Greater Sydney and Newcastle every week to monitor COVID-19 concentrations. The results are published in weekly reports. During week 38, gene concentrations per 1,000 people have been low in all the catchment areas.

New Zealand

Last data update: 14 September

The dashboard was launched in 2020 to help the public monitor potential COVID-19 risks in their local areas using easy-to-understand data visualisations. Starting in August 2024, data are updated every two weeks, and information on variants is updated at least once a month. Sites and frequency may change over time. Beginning in July 2024, variant analysis is conducted across all wastewater sites.
SARS-CoV-2: SARS-CoV-2 concentrations in wastewater are being monitored at both national and regional levels. In week 37, 11 sites covering 45% of the population tested positive for SARS-CoV-2. In week 35, the dominant strain was the NB.1.8.1, which accounted for 55.3% of cases. This was followed by the XFG strain at 21%. Trends are visually represented on a map.

Figure 25. Average SARS-CoV-2 genome copies detected per person per day in wastewater for New Zealand, along with the reported case numbers.



Source: ESR Wastewater Surveillance.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

NORTH AMERICA

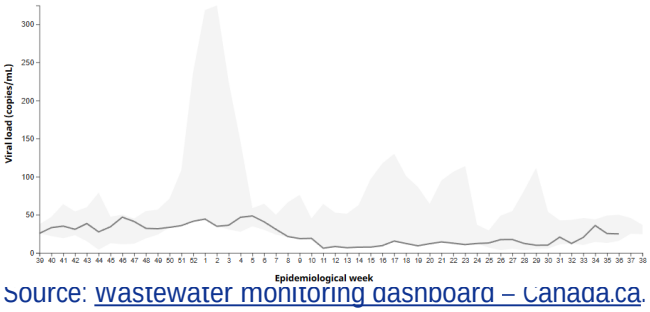
Canada

Last data update: 14 September 2025

The dashboard provides information on the levels of COVID-19 and its variants detected in the wastewater from various communities across Canada. Currently, 44 sites submit sewage samples for processing, representing 31.2% of the population. Additionally, the dashboard includes links to 23 Canadian wastewater COVID-19 surveillance dashboards and websites, which are managed by provincial, territorial, and academic partners across the country.

SARS-CoV-2: In week 37, the COVID-19 activity level in Canada was classified as moderate. Out of the 44 tested sites, 12 showed a high activity level, 15 showed a moderate activity level, 5 showed a low activity level, 6 had no detection and 1 was not available. At the end of July 2025 the dominant variant identified was XFG (63.8%) followed by NB.1.8.1 (16.5%).

Figure 26. Trend graph of 7-day rolling average of COVID-19 viral load in wastewater.



Source: Wastewater monitoring dashboard – Canada.ca.

United States

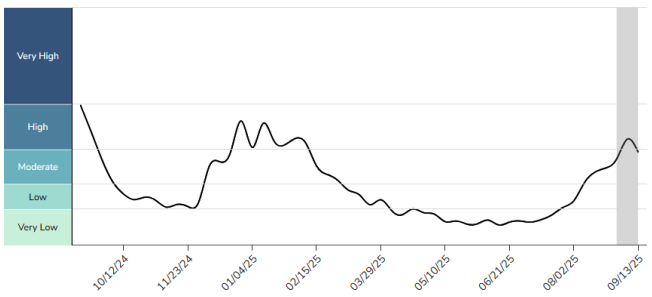
Last data update: 13 September 2025

The National Wastewater Surveillance System (NWSS) collects and analyses wastewater samples on a weekly basis. This monitoring is conducted in all 50 states, 7 territories, and some tribal communities. Over the past two months, 1,341 sampling sites have reported data to the NWSS, representing approximately 44% of the U.S. population.

SARS-CoV-2: During week 37, the national viral activity level for COVID-19 in wastewater was high.

The predominant variant during this week was XFG, which accounted for 77% of the total abundance.

Figure 27. COVID-19 Wastewater Viral Activity Level Over 1 Year Time.



Source: COVID-19 Wastewater Data – National Trends | NWSS | CDC.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox surveillance

This section provides a comprehensive overview of recent mpox outbreaks with a focus on wastewater surveillance and related news from around the world. On September 5, 2025, the World Health Organization (WHO) officially declared the end of the global Mpox emergency, following sustained reductions in case numbers and mortality rates, particularly in the Democratic Republic of Congo, Burundi, Sierra Leone, and Uganda. Despite this milestone, Mpox remains a significant public health concern, especially in Africa, which continues to account for over 90% of reported cases. According to the WHO's latest situation report (dated September 19), a total of 3,780 confirmed cases and 15 deaths were recorded globally in August. Both Clade I and Clade IIb strains remain in circulation. Mpox transmission in Europe remains low but persistent. Most cases continue to involve Clade IIb. In recent months, several EU/EEA countries—including Germany, France, Belgium, Sweden, and Ireland—have reported isolated Clade I cases, all linked to travel from endemic regions in Africa. The UK confirmed 16 Clade I cases and over 580 Clade IIb cases since 2023, with no evidence of sustained community transmission of Clade I. Berlin experienced a rise in Clade IIb cases earlier in the year, prompting renewed public health outreach. Overall, although Europe has seen some cases, the situation remains under control due to ongoing surveillance and preventive measures.

Mpox wastewater surveillance

This section provides information on mpox wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities.

EU Countries

Spain

Within the framework of the Spanish Wastewater Surveillance Programme, in December 2024 and January 2025, 10 different sampling points were assessed and 100% of them tested negative for mpox by RT-qPCR assays. The team is still working on returning to take weekly samples on 66 sampling points, due to some funding matters, and assess again routine SARS-CoV-2 samples, adding mpox, enterovirus and poliovirus to the monitored agents of interest list.

Non-EU Countries

Canada

The wastewater dashboard summarises the consistent detection of mpox virus signals in wastewater samples from selected cities across Canada. Signal interpretation is based on data from only four weeks of testing. Recent data from week 36 indicate no detection, or detection for just one week in the past four weeks, across 5 cities. In contrast, there was intermittent detection—defined as detection of mpox for at least two weeks within the past four weeks—in one city.

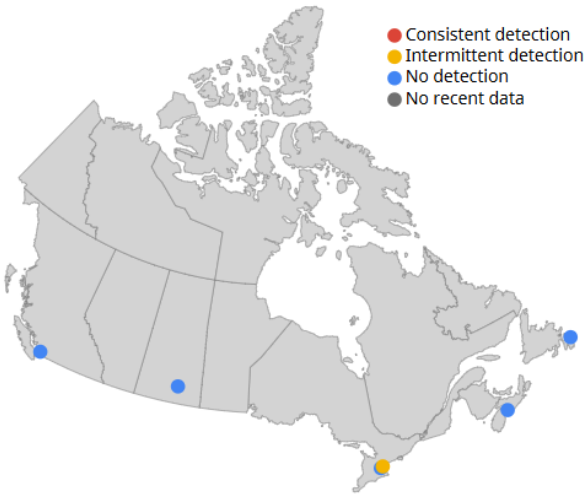
MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox surveillance

Canada

Figure 28. Recent mpox detection in wastewater in cities across Canada.



Source: Mpox activity - Wastewater monitoring dashboard — Canada.ca.

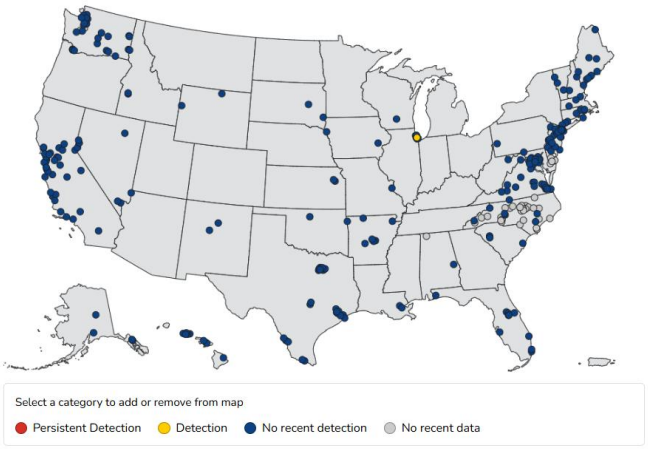
United States

The CDC's National Wastewater Surveillance System (NWSS) provides a public health framework to monitor infectious diseases through wastewater across the country. Wastewater testing for the mpox virus differs by location. Some locations test for specific clades, focusing on either clade II or clade I, while others use methods that detect both clades without distinguishing between them.

Communities are voluntarily monitoring the presence of mpox in wastewater on a weekly basis. During the four-week period from August 17, 2025 - September 13, 2025, 4 out of 531 wastewater sites (<1%) that tested for mpox reported the presence of the virus. Specifically, 3 sites indicated the detection of clade II.

The remaining sites either reported no mpox detection in wastewater (486 sites) or had no recent data available (44 sites).

Figure 29. Mpox virus detection in wastewater in the past 4 weeks, United States.



Source: U.S. Mpox Wastewater Data | National Wastewater Surveillance System | CDC.

Mpox news

15.08.2025
Africa to integrate cholera and mpox response

03.09.2025
Situation Report: Mpox Multi-Country Outbreak - Region of the Americas - 3 September 2025 - PAHO/WHO | Pan American Health Organization

05.09.2025
Mpox no longer international health emergency but remains concern in Africa | Health News | Al Jazeera

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EU Wastewater Observatory for Public Health

Mpox news

11.09.2025

Mpox outbreak: epidemiological overview, 11 September 2025 - GOV.UK

15.09.2025

Detection of Monkeypox Virus Clade Ib DNA in Wastewater Solids at Wastewater Treatment Plants, United States

15.09.2025

Mpox in the United States and Around the World: Current Situation | Mpox | CDC

16.09.2025

Mpox in 2025: What we know, what we have and what needs to happen next

17.092025

Introduction of Mpox Virus Clade Ib into the Republic of the Congo

19.09.2025

Multi-country outbreak of mpox, External situation report #58 - 19 September 2025 - World | ReliefWeb

26.09.2025

Global transmission characteristics of mpox outbreaks: a systematic review and meta-analysis

Avian Flu (H5N1) surveillance

In addition to monitoring avian flu trends through wastewater surveillance data from national dashboards and National Health Authorities' websites, this section provides a comprehensive overview of recent Avian flu news.

Avian Flu in Wastewater: Since the beginning of 2025, avian influenza A(H5N1) has been increasingly detected in wastewater in various regions, especially in the United States. In Europe and worldwide, there has been a continued rise in outbreaks among wild birds and mammals, with new cases emerging even in remote areas, such as the sub-Antarctic islands.

Global Situation: Avian influenza A(H5N1) remains a significant global concern, with widespread outbreaks continuing among wild birds and poultry across multiple continents. According to the WHO, the global public health risk from H5N1 remains low, though occupational exposure risk is considered low to moderate depending on hygiene and biosecurity measures. Human infections are still primarily linked to direct contact with infected poultry or contaminated environments. There is no evidence of sustained human-to-human transmission.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Avian Flu News

02.08.2025
Surveillance on California dairy farms reveals multiple sources of H5N1 transmission
02.09.2025
Assessing the spatial risk of wild birds in avian influenza transmission using global risk score
18.09.2025
Detection of H5N1 clade 2.3.4.4b in external quality assessment programmes: a pandemic preparedness initiative
22.09.2025
Poland reports bird flu outbreak on farms in north, WOAH says | Reuters
24.09.2025
avian-influenza-june-september-2025.pdf
25.09.2025
Avian influenza overview June–September 2025 | EFSA
25.09.2025
Understanding avian influenza mortality
27.09.2025
DHS S&T Avian Influenza Report Signals Rising Global Health Security Risks
28.09.2025
Bird flu (avian influenza): latest situation in England - GOV.UK

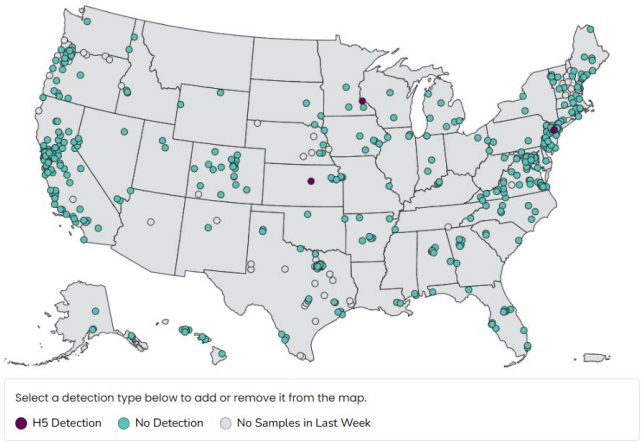
Avian flu wastewater surveillance

This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

NON-EU countries

United States. The CDC receives weekly data on the avian influenza A(H5) virus from a contract partner, an academic group, and several state health departments. Wastewater sampling and testing are conducted one or more times each week.
During week 37, avian influenza A(H5) results were reported from 404 sites. Of these, 3 sites (0.7%) across 3 states reported H5 detections, while 401 sites reported no detections.

Figure 30. Site-level data for avian influenza A(H5) viral RNA detections in wastewater.



Source: Wastewater Data for Avian Influenza A(H5) | National Wastewater Surveillance System | CDC.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus surveillance

This section provides a summary of latest detected news on poliovirus to keep informed about the most current developments.

Europe: Between September and December 2024,, vaccine-derived poliovirus type 2 (cVDPV2) was detected in wastewater samples across multiple European countries, including Germany, Poland, Spain, Finland, and the United Kingdom. These detections are part of ongoing environmental surveillance efforts and no human polio cases have been reported in association with them. These detections primarily involve circulating vaccine-derived poliovirus type 2 (cVDPV2) genetically linked to a lineage first identified in Nigeria in 2020.

As of October 2025, no human cases of polio or poliovirus infections have been reported in association with these environmental detections in the EU/EEA, and the region remains free from the disease. However, the presence of poliovirus in wastewater highlights the need for continued vigilance and preventive measures, especially in under-vaccinated communities.

The European Centre for Disease Prevention and Control (ECDC) emphasises the importance of maintaining robust surveillance systems, including acute flaccid paralysis (AFP) monitoring and environmental surveillance, to rapidly detect and respond to any circulation of poliovirus.

Global update: Polio remains a global health concern, with wild poliovirus type 1 (WPV1) still endemic in Pakistan and Afghanistan, where 13 cases have been reported so far in 2025. Circulating vaccine-derived poliovirus type 2 (cVDPV2) continues to spread across 35 countries, with detections increasingly found in wastewater samples in regions including Europe, Africa, and Asia.

Poliovirus News

28.07.2025

Statement of the Forty-second meeting of the Polio IHR Emergency Committee – WHO

07.08.2025

Polio virus detected in Jerusalem, central Israel wastewater, Health Ministry confirms

31.08.2025

Poliomyelitis dynamics with fractional order derivatives and deep neural networks | Scientific Reports

01.09.2025

Towards ultra-sensitive and rapid near-source wastewater-based epidemiology | Nature Communications

05.09.2025

Intestinal mucosal immune responses to novel oral poliovirus vaccine type 2 in healthy newborns | Clinical Infectious Diseases | Oxford Academic

12.09.2025

Exploring the potential of hydro alcoholic crude extract of beeswax as antibacterial antifungal antiviral antiinflammatory and antioxidant agent | Scientific Reports

Poliovirus Wastewater Surveillance

This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus Wastewater Surveillance

EU countries

Belgium. Belgium has started monitoring poliovirus in January 2024. The surveillance program involves collecting wastewater samples every four weeks from the influent of 6 wastewater treatment plants. They do not plan to publish poliovirus data in a report.

Finland. Finland has been monitoring poliovirus in wastewater since the 1960s. Samples are collected biweekly in the Helsinki capital region and monthly in four other cities, covering approximately 30% of the Finnish population. At the moment, no data is being reported.

Ireland. Since February 2023, samples from the largest wastewater catchment area in Ireland have been tested every two weeks for poliovirus 2 (PV2). Currently, the focus is on PV2, based upon recent detections elsewhere of cVDPV2. Additional catchment areas and poliovirus types may be added over the coming months.

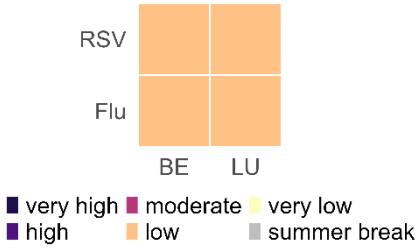
Slovenia. In 2024, Slovenia began monitoring for poliovirus in wastewater to detect any potential reintroduction of the virus into the country. This monitoring is conducted every three months at 16 WWTPs from all statistical regions, covering 34.2% of the population. The latest monitoring results, from September 2025, showed that all samples were negative for the virus.

Influenza, RSV and other viruses' surveillance

EU countries

This section provides information on RSV, influenza and other viruses wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Figure 30 illustrates the comparative viral loads of Influenza and RSV across various EU countries and includes only those countries for which indications on viral loads are provided.

Figure 31. Viral loads for Influenza and RSV across EU countries..



Source: EC-JRC

Influenza

EU countries

Austria (regional data). Since October 2022, the influenza virus has been monitored weekly in the city of Vienna. The virus load is depicted in the graphics updated to week 37 and the data, including previous seasons, is available for download as an open governance dataset.

MONTHLY BULLETIN

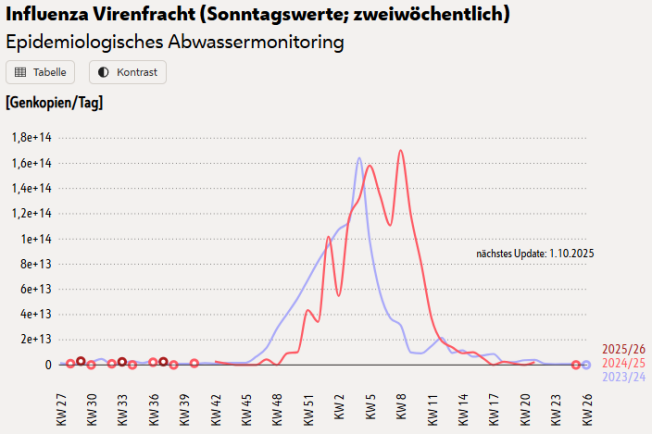
EU Wastewater Observatory for Public Health

Influenza

EU countries

Austria

Figure 32. Influenza virus load in Vienna.

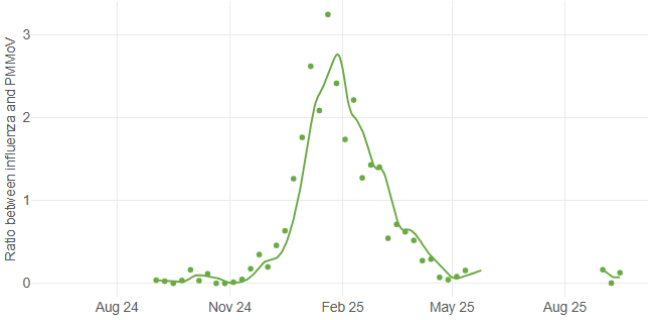


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium has been monitoring the presence of influenza A and B since the beginning of this year. The surveillance program includes collecting wastewater samples once a week from the influent of
30 wastewater treatment plants. During week 38, the Influenza concentration was at a low level.

Figure 33. Ration between Influenza and PMMoV.

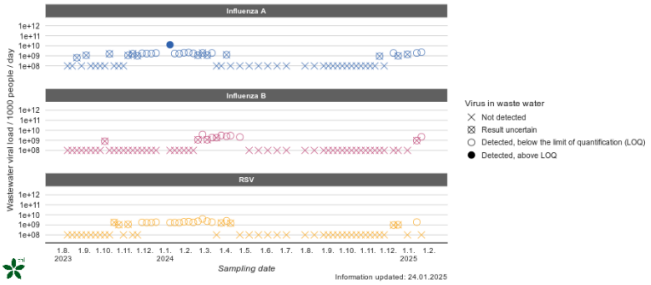


Source: Wastewater Sciensano.

Finland

The results for influenza A and influenza B, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 8 of these locations have been updated up to week 37 and can be downloaded in CSV format.

Figure 34. Influenza A and B viral loads in wastewater (top and middle panels), Helsinki (information updated: 24.01.2025).



Source: Finland

EU countries

Germany

Influenza A and Influenza B viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. For calendar week 38, data from 58 WWTPs were available. The average total viral load for Influenza A and B decreased in the past 4 weeks.

Additionally, monitoring for Influenza A and B in wastewater has been conducted in the German state of Bavaria since the beginning of 2024. Trends for Influenza are reported weekly on the local dashboard. During week 39, the viral load showed a decrease in 28% of the monitored locations, an increase in 67%, and remained stable in 5% of the locations.

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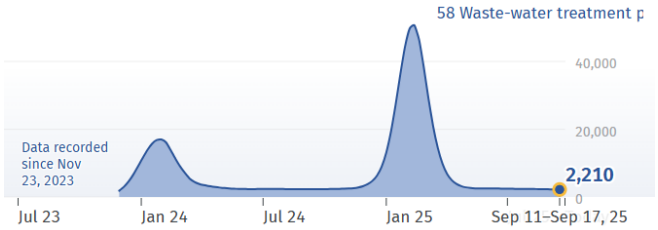
EU Wastewater Observatory for Public Health

Influenza

EU countries

Germany

Figure 35. National influenza viral load in wastewater (gene copies per liter).

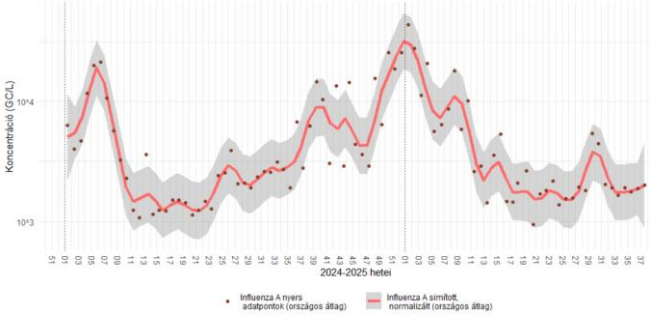


Source: Infektionsradar.

Hungary

The Influenza A virus has been detected in wastewater since January 2023. Starting from week 48 of 2024, the results for influenza A were supplemented with the amount of influenza B genetic material detected. The sampling locations are the same as those used for SARS-CoV-2. In week 37, the national average concentration of influenza A and Influenza B genetic material are stagnating.

Figure 36. National, population-weighted average of Influenza A copy number measured in wastewater.

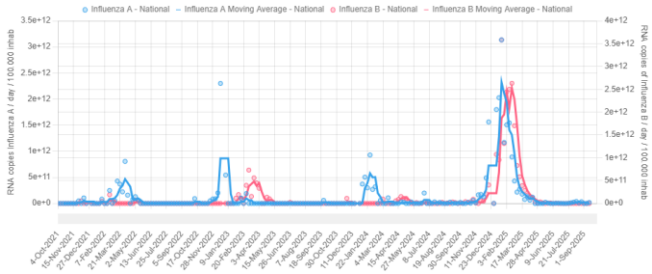


Source: National Center for Public Health and Pharmacy - 2025. Week 25 wastewater test results.

Luxembourg

The dashboard monitors the levels of Influenza A and B in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since October 2021, with updates every two weeks. As of week 38, the loads of Influenza A and B are low.

Figure 37. Influenza A/B national levels (RNA copies/day/100.00inhab).

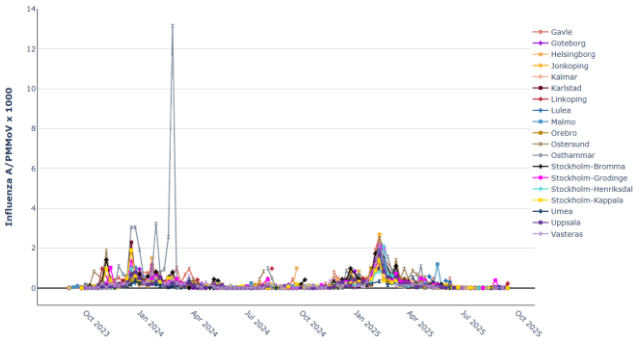


Source: Influenza – MicrObs.

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for influenza A and B viruses from 24 sites across Sweden, covering 43% of the population. The results are available up to week 38.

Figure 38. PMMoV normalised Influenza A content.



Source: Amount of influenza virus in wastewater (SLU) | The Swedish Pathogens Portal.

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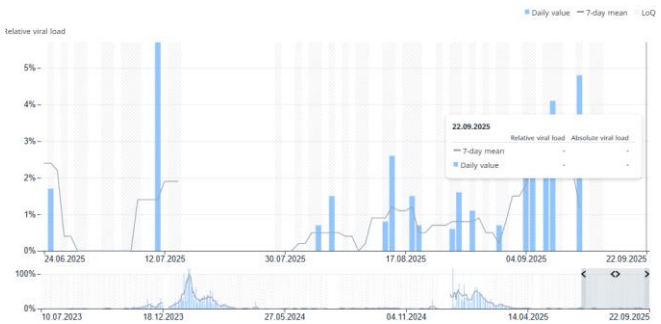
Influenza

Non-EU countries

Switzerland

Influenza A and B have been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring will be conducted at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein. In week 38, the trends for Influenza A and B indicated a stagnation across all regions.

Figure 39. Development over time of influenza detections.



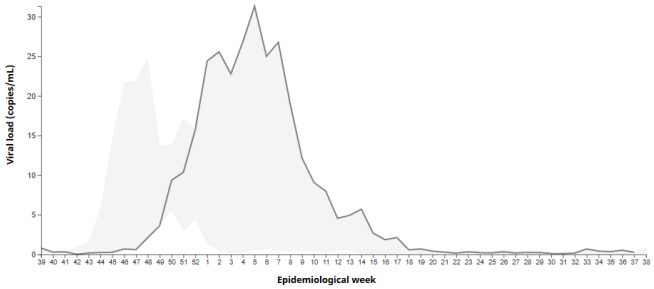
Source: IDD - Infectious Diseases Dashboard.

Canada

The dashboard provides information about the levels of Influenza A and B found in the wastewater from various communities across Canada.

In week 36, 19 sites are submitting sewage samples for analysis, which represents 29.2% of the Canadian population. The activity level of Influenza A was classified as low. Among the 19 tested sites, 3 sites showed a high activity level, 5 showed a moderate activity level, 2 showed a low activity level and 9 reported no detection. For Influenza B, the activity level was classified as low. Out of the 19 tested sites, 1 showed a low activity level, and 18 had no detection.

Figure 40. National average of Flu A viral load (copies/ml) in wastewater.



Source: Wastewater monitoring dashboard – Canada.ca.

Japan

In October 2024, Komatsu City began analysing and sharing information about the influenza A virus, providing updates 2-3 times a week. Influenza virus analysis is conducted from October to March.

No updates are expected until next autumn.

United States

The CDC and public health partners are continuing to investigate whether the changing levels of influenza A virus in wastewater are related to seasonal influenza, the avian influenza A (H5) outbreak in animals, or both. Wastewater data cannot determine the specific source of influenza A viruses.

In week 37, the activity level of influenza A in wastewater across the country was very low.

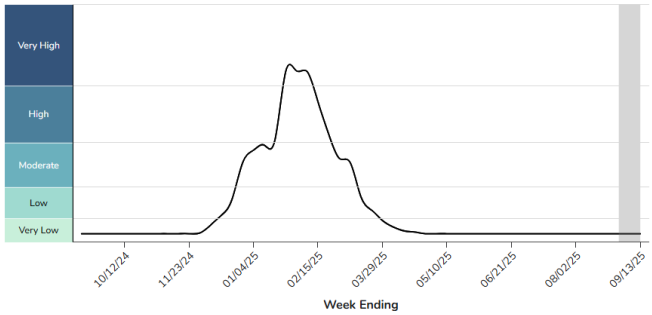
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EU Wastewater Observatory for Public Health

Influenza

United States

Figure 41. National trends of wastewater viral activity levels of influenza A.



Influenza A Wastewater Data – National Trends | NWSS | CDC.

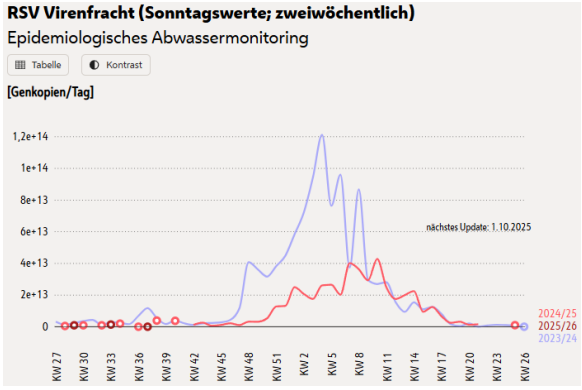
RSV

EU countries

Austria

(regional data). Since October 2022, the RSV has been monitored on a weekly basis in the city of Vienna. The virus load is depicted in the graphics updated to week 40, and the data, including previous seasons, is available for download as an open governance dataset.

Figure 42. RSV virus load in Vienna.

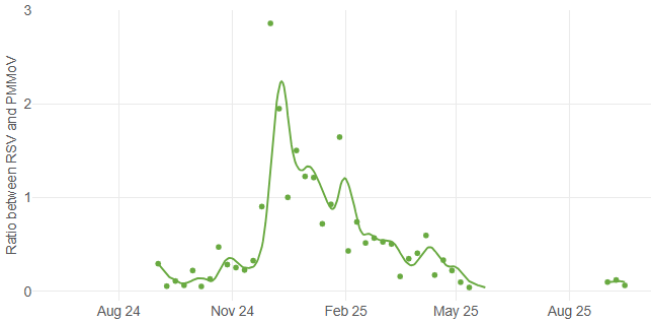


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium started monitor RSV since the beginning of this year. The surveillance program involve the collection of wastewater samples once a week from the influent of 30 wastewater treatment plants. During week 38, the RSV concentration was at a low level.

Figure 43. Ratio between RSV and PMMoV.

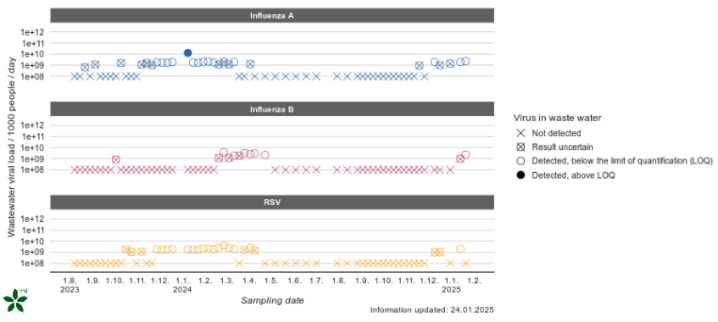


Source: wastewater sciensano

Finland

The results RSV, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 8 of these locations have been updated up to week 37 and can be downloaded in CSV format.

Figure 44. RSV viral load in wastewater (bottom panel), Helsinki (Number of residents within the network: 860,000). (information updated: 24.01.2025).



Source: Influenza A, influenza B, and RSV in wastewater – THL.

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EU Wastewater Observatory for Public Health

RSV

EU countries

Germany

RSV viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. During reporting week 38, data from 36 WWTPs were available. The average viral load for RSV increased in the past 4 week.

Since the beginning of 2024, wastewater monitoring for RSV has been conducted in Bavaria, Germany. Trends for RSV are reported weekly on the local dashboard. In week 38, the viral load decreased in 44.4% of the monitored locations, increased in 50%, and remained stable in 5.6%.

Figure 45. National RSV viral load in the wastewater (no. of gene copies per litre).

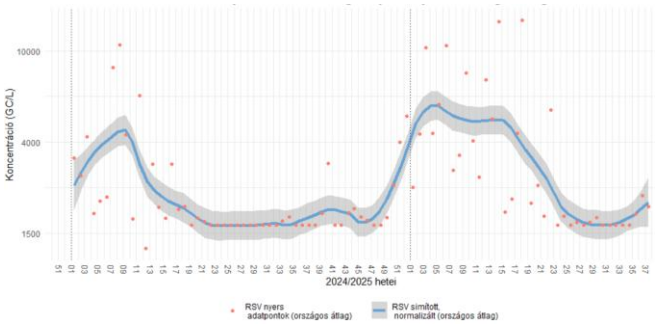


Source: Infection Radar.

Hungary

RSV virus has been detected in wastewater since January 2023. The sampling locations are the same as those for SARS-CoV-2. During week 37, the national average concentration of RSV was slightly increasing and was above the detection limit in two service areas.

Figure 46. National, population-weighted average of RSV copy number measured in wastewater.

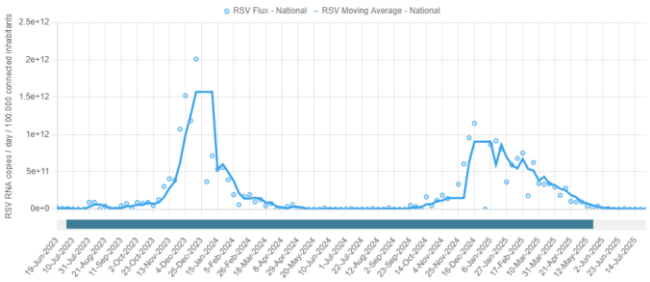


Source: National Center for Public Health and Pharmacy - 2025. Week 25 wastewater test results.

Luxembourg

The dashboard monitors the levels of RSV in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since June 2023, with updates every two weeks. As of week 38, RSV levels are low.

Figure 47. RSV national levels (RNA copies/day/100.00inhab).



Source: RSV – MicrObs.

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EU Wastewater Observatory for Public Health

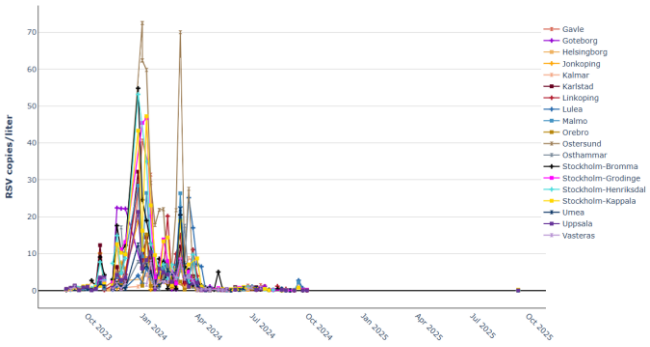
RSV

EU countries

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for RSV from different areas of Sweden. The results are available up to week 38 for 19 cities.

Figure 48. RSV concentration in copies/liter.



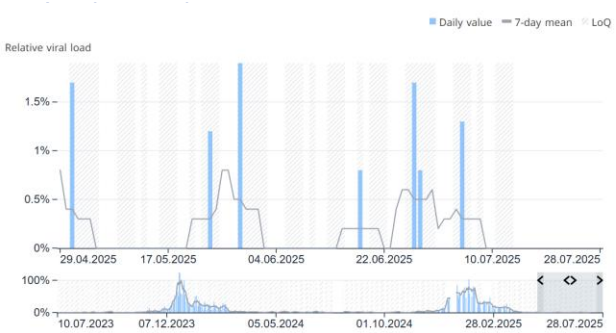
Source: Amount of Respiratory Syncytial Virus (RSV) in wastewater (SEEC-SLU) | The Swedish Pathogens Portal.

Non-EU countries

Switzerland

RSV has been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring takes place at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein. As of week 38, the RSV trends have been stagnant across all 11 regions.

Figure 49. RSV relative viral load for a selected



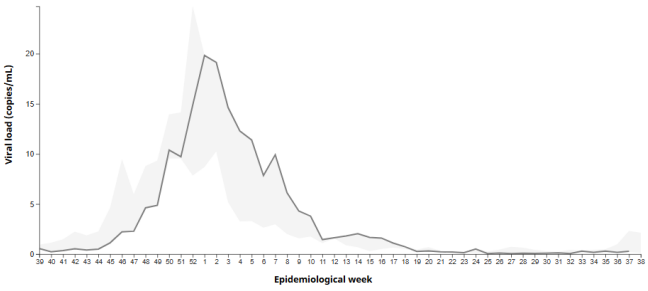
Source: IDD - Infectious Diseases Dashboard.

Canada

The dashboard presents data on the levels of RSV detected in the wastewater of various communities across Canada. Currently, 19 sites are submitting sewage samples for analysis, representing 29.2% of the Canadian population.

In week 39, the RSV activity level was classified as low. Of those sites, 2 showed a high activity level, 3 showed a moderate activity level, 4 showed a low activity level and 8 had no detection.

Figure 50. National average of RSV viral load (copies/ml) in wastewater.



Source: Wastewater monitoring dashboard – Canada.ca.

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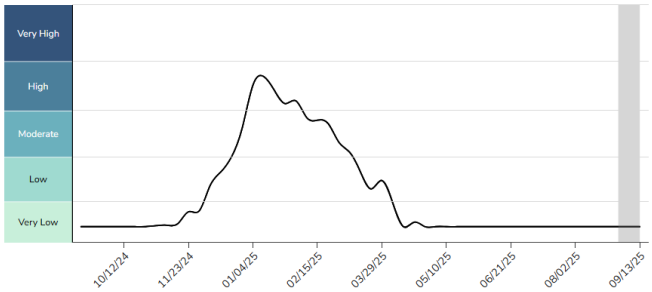
RSV

Non-EU countries

United States

CDC's National Wastewater Surveillance System (NWSS) provides the public health infrastructure to monitor RSV through wastewater across the country. During week 37, the level of viral activity for RSV in the wastewater was very low on a national level.

Figure 51. National trends of wastewater viral activity levels of RSV.



Source: RSV Wastewater Data – National Trends | NWSS | CDC.

Enteric Viruses

EU countries

Luxembourg

Norovirus and Enterovirus have been monitored in Luxembourg from March 2020 to July 2024. Wastewater levels were reported from two cantons, covering approximately 52% of the population through four treatment plants. Latest sampling date: 15.07.2024.

Sweden

Norovirus and Enterovirus have been monitored in Luxembourg from March 2020 to July 2024. Wastewater levels were reported from two cantons, covering approximately 52% of the population through four treatment plants. Latest sampling date: 15.07.2024.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

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